Orchestrate or Upgrade? When a Lower-Tier Model Delivers 98% Quality at 23% Cost

Youssef Hariri

DeeWaanAI · osf.io/mjb8x · 2026-08-28

Abstract

Organizations deploying large language models (LLMs) for knowledge work face a budget choice: pay for a frontier model, or orchestrate cheaper models into a multi-step pipeline. This thesis answers that choice with three pre-registered studies that escalate a single question.

Study 1 — Does structure substitute for tier? In a 2×2×2 factorial across five LLM families and 19 business problems (1,900 blinded generation–judgment runs), adding a three-step team structure lifts a lower-tier model to **98.8% of frontier quality** at 88% of the cost (T-EGR = 1.49), and structure helps cheaper models significantly more than frontier ones ($d = -0.227$, $p = 0.015$). Yet a *single* lower-tier call already captures 98% of frontier quality at just 23% of the cost (T-EGR = 6.15) — the highest-return configuration tested — so the team's added structure buys only a marginal quality gain for about 3.8× the spend. Frontier teams are the worst dollar (3.5× the cost for negligible gain, T-EGR = 0.27). A pre-registered replication on 30 fresh problems and 10 families confirms the core economics out-of-sample; three secondary hypotheses — a marketing-domain exception, a capability-gap correlation, and team-driven variance reduction — failed their pre-registered gates and are not recommended.

Study 2 — Why substitution stops short. A $0.002 refinement pass recovers up to 64% of the frontier gap but never achieves paired superiority. Qualitative coding ($\kappa = 0.824$) locates the ceiling: orchestration buys *coverage* (more structure, more length) while the frontier wins on *density* (compressed insight, zero boilerplate) — a gap that add-only pipelines cannot close.

Study 3 — Who should design the pipeline? When models auto-design their own workflows, authorship barely matters: self-designed and frontier-designed pipelines are statistically indistinguishable in four of five families, and both beat our frozen human-designed instrument. But orchestration economics are family-specific — at 0.24–3.5× solo cost, cheap auto-designed pipelines beat the frontier-solo baseline for Kimi ($d = +0.85$ on novelty, the only family to survive multiplicity control) and lose decisively for Claude ($d = -3.15$). A pre-registered replication on 30 fresh problems finds the pattern transports only at its extremes (Kimi $d = +1.25$; Claude $d = -1.90$), so routing must be *measured per deployment, not looked up.*

For business decision-makers: use a *single* lower-tier call by default — the highest return. Add a *team* (structure) only when the binding gap is *coverage* (breadth, length, completeness) and its roughly four-fold cost is justified. Reserve the *frontier* for *density*-critical work — compressed insight with zero boilerplate —

the one thing orchestration cannot manufacture. Never combine frontier and team (the worst dollar in the study). And because these economics are family-specific, *measure* efficiency (T-EGR) on your own models and problems rather than trusting any lookup table.

Keywords: Large Language Models, Multi-Agent Systems, LLM Orchestration, AI Cost Optimization, Pipeline Architecture, LLM-as-a-Judge, Resource Allocation, Pre-registered Experiment, Cost-Effectiveness Analysis, Business AI Strategy, T-EGR, Model Tier Substitution.

A practitioner-oriented executive summary with budget-allocation rules appears at the end of the paper (Appendix A).

List of Abbreviations

Abbr	Definition
BF_{10}	Bayes factor in favor of H_1 over H_0
C-S-N	lower-tier solo, no persona
C-S-P	lower-tier solo, persona
C-T-N	lower-tier team, no persona
C-T-P	lower-tier team, persona
F-S-N	frontier solo, no persona
F-S-P	frontier solo, persona
F-T-N	frontier team, no persona
F-T-P	frontier team, persona
EGR	Efficiency Gain Ratio
FDR	false discovery rate
FLOPs	floating-point operations
FSN	frontier solo, no persona (baseline)
ICER	incremental cost–effectiveness ratio
IRR	inter-rater reliability
JZS	Jeffreys–Zellner–Siow (Bayes factor prior)
LLM	large language model
LMM	linear mixed model
LOO	leave-one-out
OBF	O'Brien–Fleming (spending function)
SESOI	smallest effect size of interest
T-EGR	Tiered Efficiency Gain Ratio
TOST	two one-sided tests (equivalence)

1 Introduction

Recent work has moved from single-model inference to the orchestration of multiple language-model calls: refinement pipelines chain builder, critic, and refiner steps into structured workflows (Wei et al., 2022; Shinn et al., 2023; Yao et al., 2023); agent teams assign social roles and have models debate toward an answer (Liang et al., 2024); and panels of judges aggregate verdicts across models (Zheng et al., 2023; Verga et al., 2024).

Across these designs runs an implicit economic claim — that *structure can substitute for model tier*. A practitioner who cannot afford a frontier model, the argument runs, can reach comparable output quality by composing cheaper models into an orchestrated system. If true, the claim reshapes how organizations price, budget, and architect generative-AI products. If false, orchestration is at best a marginal convenience and at worst a cost multiplier that never reaches frontier quality.

Prior empirical work on generative AI is encouraging but silent on this specific claim. Controlled productivity studies show that a single generative assistant raises output quality and reduces task time (Noy and Zhang, 2023; Brynjolfsson et al., 2023), yet they treat the model as a given, single-agent capability. A field experiment with consultants documented a "jagged technological frontier": large gains inside the model's capability boundary, large losses outside it (Dell'Acqua et al., 2023). Neither study varies the structure around the model. Multi-agent and ensemble work shows that sampling, debate, and tree search can improve accuracy (Wang et al., 2023; Liang et al., 2024; Yao et al., 2023), but the gains are reported without cost-matched comparison to a stronger single model. A nascent theory of test-time compute predicts that, for fixed problem difficulty, more inference compute on a smaller model can outperform less inference compute on a larger one (Snell et al., 2024), but its predictions are in units of FLOPs, not dollars, and only on verifiable tasks.

Four gaps follow. First, no factorial, cost-matched comparison of orchestrated lower-tier models versus frontier solo exists on open-ended tasks. Second, no mechanistic account explains *why* orchestration succeeds or fails. Third, no economic metric is robust to the price-assumption choices that inevitably shape per-run costing. Fourth, orchestration studies almost invariably freeze a *human-designed* pipeline: whether models can design their own workflows, and whether the resulting cost–quality economics transport across model families and fresh problems, is largely untested — leaving practitioners unable to tell whether a winning configuration can be copied or must be re-measured.

This paper addresses all four. We run a pre-registered $2 \times 2 \times 2$ (tier $\times$ structure $\times$ persona) factorial experiment over five model families. The third factor — a fixed cultural persona applied to each pipeline role, carried from our prior work on persona economics (Hariri, 2026b) — is a secondary, boundary question rather than part of the core tier–structure claim; it produces a large negative effect, with personas raising cost while degrading quality (Section 4.1.2). Nineteen confirmatory business problems are drawn from a 30-problem domain-stratified universe, with 11 problems reserved for feasibility gating per the pre-registered protocol — a separation that keeps feasibility probes from anchoring confirmatory inference (Albers and Lakens, 2018). The experiment produces 1,900 generation–judgment runs, measured by a blinded three-judge panel from families disjoint from every generator. Cost is stored as tokens and re-priced through a frozen price vector under three blended input/output ratios, so every reported dollar is reproducible and re-priceable. A second study applies a cheaper, add-only refinement pass to the orchestrated outputs and codes the qualitative differences between original, refined, and frontier answers, testing the mechanistic prediction that orchestration buys coverage while the frontier wins on density. A third study unfreezes the pipeline itself: across the same five families, models auto-design their own workflows, which we compare against frontier-designed and our frozen human-designed instruments, and we test whether the resulting cost–quality economics transport to 30 fresh problems in a pre-registered replication.

Six contributions follow:

1. **The first cost-matched, pre-registered factorial** testing whether orchestration

structure substitutes for model tier on open-ended business tasks — with negative superiority results reported as primary findings, not buried.
2. **A new economic metric, T-EGR** (quality per dollar, relative to the frontier baseline) and a cost–quality Pareto frontier shown to be robust to price assumptions.
3. **A mechanistic explanation** of why substitution stops short: orchestration buys *coverage* while the frontier wins on *density* — a gap a low-cost refinement pass narrows but never closes.
4. **A pipeline-authorship finding**: who designs the workflow barely matters, but the economics are family-specific, so orchestration must be *measured per deployment, not looked up*.
5. **Practitioner contingency rules** mapping each budget tier and task type to the pipeline architecture with the highest expected return.
6. **External validity — and its boundary**: across two confirmatory replications on fresh problems and a widened family universe, the core cost–quality economics generalize, while the contingent nuances do not, identifying which orchestration findings practitioners can rely on.

The paper proceeds as follows. Section 2 reviews related literature. Section 3 describes the pre-registered design, apparatus, and analysis, including an AI-assisted research disclosure (Section 3.9). Section 4 presents the three studies as a single buyer's decision journey: Study 1 asks whether structure substitutes for model tier, Study 2 asks why substitution stops short of the frontier, and Study 3 asks who should design the pipeline. Section 5 discusses theoretical and practical implications; Section 6 translates the findings into budget rules and an operating model for practitioners. Section 7 acknowledges limitations. Section 8 outlines future directions. Section 9 concludes.

2 Literature Review

This review synthesizes four streams of research: (i) measured productivity effects of generative AI, (ii) multi-agent and ensemble generation methods, (iii) LLM-as-a-judge evaluation methodology, and (iv) open-science methods for economic measurement. We then integrate these streams with our own prior work on personas and efficiency metrics to articulate the research gap that motivates this study.

2.1 Productivity effects of generative AI

2.1.1 Measured gains in knowledge work

Three controlled studies establish that generative AI raises knowledge-work productivity, but they also reveal that the gains are conditional and bounded. Noy and Zhang (2023) randomized 444 mid-level professionals to ChatGPT-assisted and control arms on professional writing tasks: assistance reduced completion time by 0.8 standard deviations and raised output quality by 0.4 standard deviations, with the largest quality gains among initially lower-performing workers — a compression of the performance distribution. Brynjolfsson et al. (2023) observed 5,179 customer-support agents using a generative assistant in the field and found a 14% average productivity increase, concentrated among novice and low-skilled workers (+34%), consistent with the model diffusing behaviors of high-ability agents across the workforce.

A critical limitation unites these studies: both treat the model as a *given, single-agent capability*. The unit of analysis is one model call, not an orchestrated system. The professional-writing and customer-support tasks they study are also relatively closed-form: the desired output is well-specified (a support response, a draft document), and quality can be partially verified against ground truth. Open-ended strategic tasks — the domain of this paper — have no verifiable correct answer, which changes both the nature of the "frontier" and the applicability of the productivity gains documented above.

2.1.2 The jagged frontier and its boundary

The boundary of these gains is the central caution. Dell'Acqua et al. (2023) ran a field experiment with 758 consultants on realistic professional problems and documented a "jagged technological frontier": inside the model's capability boundary, consultants completed 12.2% more tasks, 25.1% faster, with 40% higher quality; outside it, consultants using the model were 19 percentage points less likely to produce correct solutions than controls. Two productive behavioral strategies emerged ("centaurs" dividing work at the boundary, "cyborgs" interleaving with the model), but neither changed the location of the boundary itself.

The jagged frontier concept is especially consequential for open-ended tasks. In closed-form domains (arithmetic, code, support scripts), the boundary is sharp: a solution is either correct or not. In open-ended business strategy, the boundary is diffuse — there is no single correct answer, only better or worse responses. This diffuseness creates a fundamental ambiguity: when a cheaper model produces a plausible but suboptimal strategy, the user may not know they are "outside the frontier" until the solution fails in deployment. The open question this paper addresses is whether *orchestration structure* can shift the effective capability frontier for these open-ended tasks — and at what cost.

2.2 Multi-agent systems, reasoning, and inference-time compute

2.2.1 From chain-of-thought to structured reasoning

The modern reasoning literature begins with the observation that language models solve complex problems better when they generate intermediate reasoning steps. Wei et al. (2022) introduced chain-of-thought (CoT) prompting: appending exemplars that show step-by-step reasoning before the final answer elicited comparable reasoning in the model, dramatically improving performance on arithmetic, commonsense, and symbolic reasoning tasks. The CoT paradigm reframed language-model inference from single-shot generation to *structured multi-step deliberation* — a conceptual shift that enabled the orchestration methods reviewed below.

CoT's direct descendant in the accuracy–cost tradeoff space is self-consistency. Wang et al. (2023) showed that sampling a diversity of reasoning paths from a single model and taking a majority vote over final answers improves arithmetic, commonsense, and symbolic reasoning well beyond CoT decoding alone. The result establishes the general principle that parallel sampled calls can be converted into accuracy. Its direct corollary — that accuracy is bought at a multiplicative cost in calls and tokens — is rarely priced against the alternative of a single stronger model.

2.2.2 Interactive deliberation: debate, tree search, and agent loops

Moving beyond independent sampling, recent work has explored *interactive* structures in which model instances influence each other. Liang et al. (2024) showed that multiple LLM instances proposing answers, critiquing each other's responses, and refining their own over debate rounds improves factuality and reasoning on arithmetic and knowledge benchmarks, encourages divergent thinking, and reliably converges to shared answers. Yao et al. (2023) extended the reasoning structure from linear chains to branching trees: Tree of Thoughts treats each reasoning step as a node in a search graph, allowing the model to explore multiple reasoning paths, backtrack from dead ends, and select the most promising branch — deliberate problem solving rather than single-path generation.

These works demonstrate that structure (sampling, aggregation, critique loops, tree search) adds quality beyond a single call. What they do not provide is a cost-matched factorial comparison: prior multi-agent studies report quality gains without measuring dollars, and without comparing orchestrated lower-tier models against an un-orchestrated frontier model on identical problems. The substitution claim — that structure can stand in for model tier — is therefore assumed, not tested.

2.2.3 The test-time compute hypothesis

A nascent theoretical framework reframes the model-tier-versus-structure tradeoff as an allocation problem. Snell et al. (2024) demonstrated that scaling inference-time computation (more tokens, verifier-based search, iterative refinement) can be more effective than scaling model parameters for improving performance on reasoning benchmarks. Their compute-optimal curve predicts that for any fixed problem difficulty, there exists an optimal allocation between training compute (model size) and inference compute (test-time search). Below that optimum, a smaller model with more inference compute outperforms a larger model with less; above it, the larger model wins.

This framework is the closest theoretical grounding for the "orchestration economics" claim, but it remains limited in three ways that motivate our study. First, Snell et al. (2024) measure *verifiable* reasoning tasks (math, code) where a verifier can score intermediate steps; open-ended business strategy has no verifier. Second, their test-time compute is *search-based* (sampling + scoring), whereas the multi-agent literature uses *structural* compute (builder–critic–refiner pipelines, debate rounds) — the two may have different cost–quality curves. Third, and most fundamentally, their analysis is in units of FLOPs, not dollars: the practitioner facing an API bill cannot pay in FLOPs. A dollar-matched empirical test is missing.

2.2.4 Self-refinement and the limits of iterative improvement

The final reasoning structure relevant to this study is self-refinement: a model critiquing and revising its own output. Shinn et al. (2023) introduced Reflexion, in which a language agent maintains a reflective memory of past failures and uses verbal reinforcement learning to improve subsequent trials. On coding and decision-making tasks, Reflexion outperformed single-shot baselines and even standard CoT by learning from its own errors.

Self-refinement raises a hypothesis directly tested in our Study 2: if a cheaper model's initial output is below the frontier, can a low-cost refinement pass close the gap? The multi-agent and self-refinement literatures imply that iterative improvement should raise quality. But they do not tell us whether the improvement is sufficient to reach parity with

a frontier model that was not itself refined — the marginal comparison that determines substitution. Nor do they tell us *why* refinement succeeds or fails on open-ended tasks. Our qualitative coding protocol (Section 3.6.5) is designed to answer this mechanistic question.

2.3 LLM-as-a-judge evaluation methodology

Because our outcomes are open-ended business solutions, evaluation itself is a design object. Zheng et al. (2023) established LLM-as-a-judge as a scalable evaluation paradigm, showing strong agreement between strong judge models and human preferences on pairwise comparison tasks, while documenting three systematic biases that threaten validity: position bias (preferring the candidate presented first), verbosity bias (preferring longer outputs), and self-enhancement bias (preferring outputs from the same model family as the judge). Their mitigations — swapping candidate order, controlled few-shot examples, length normalization — inform our blinding and order-shuffling protocol.

The self-enhancement bias is particularly acute for multi-agent studies. If the generator and judge share model architecture, the judge may reward the generator's stylistic conventions rather than substantive quality. Verga et al. (2024) extended the single-judge design into a jury: a Panel of LLM evaluators (PoLL) composed of several smaller models from different families correlated better with human judgments than a single large judge, at lower cost, while mitigating the self-preference bias that afflicts same-family evaluation. Our evaluation framework inherits this logic directly: three judges from families disjoint from all generators, each under three frozen prompt variants, scoring blind to cell and family. Because no judge shares a family with any generator, the "persona penalty" reported in Section 4.1.2 cannot be attributed to same-family self-enhancement bias (Zheng et al., 2023).

2.4 Open-science methods and economic measurement

The fourth stream supplies the discipline this paper applies. Nosek et al. (2018) argue that preregistration distinguishes confirmatory from exploratory analysis, converting transparency into credibility; our protocols and their amendments were frozen before the corresponding computations ran and are timestamped in the OSF registration chain (Table 2), which also discloses the two exceptions. Steegen et al. (2016) show that conclusions can vary across reasonable analytic choices and recommend reporting a multiverse of analyses; our price-assumption panel (three blended input/output ratios) and pre-registered robustness battery implement this idea. Sandve et al. (2013) codify reproducible computational research — versioned scripts, recorded steps, stable environments — which our token ledger and frozen configuration files follow.

Two statistical practices complete the foundation. Albers and Lakens (2018) demonstrate that effect sizes estimated from pilots are unstable and bias follow-up power analyses, so pilot data must never anchor confirmatory claims; accordingly, our pilot data informed only feasibility gating, never the confirmatory analysis. Lakens (2013) makes the case for reporting effect sizes with confidence intervals to support cumulative science; we report Welch's d with bootstrap intervals throughout. Cohen (1960)'s κ coefficient provides our inter-rater and coder-agreement metric.

Finally, measuring the economics of inference requires separating quantity from price. Diewert (1976) formalized the decomposition of value changes into price and quantity

components in index-number theory; we apply the same logic by storing tokens (quantity) and multiplying by a frozen price vector, so every reported dollar is reproducible and repriceable. Tomlinson et al. (2024) quantify the environmental footprint of generative tasks, reinforcing that inference cost — economic and environmental — is a first-order design variable rather than an implementation detail.

2.5 Integration: from persona economics to orchestration economics

Our own prior work supplies the immediate empirical background. In a cross-architecture evaluation of cultural personas across seven large language models, persona effectiveness followed a contingency model rather than a universal rule, and the Efficiency Gain Ratio (EGR) showed that small models with personas achieved superior performance at 40–88% lower cost (Hariri, 2026b). The same line of work measured creativity with multi-judge panels over thousands of generated business solutions, establishing Novelty as the best-agreed creativity dimension (Hariri, 2025a), and showed that team composition effects are contingent on task verifiability (Hariri, 2025b).

Integrating the four streams above with this lineage yields the gap and a pre-registered hypothesis set spanning all three studies, each element grounded in the literature or in our prior work. The Study 1 labels H_1–H_5 follow the registered protocol exactly (OSF registration `osf.io/mjb8x`, Protocol 01): H_1 is the primary tier × structure interaction; H_2, H_3, and H_4 are the tier, structure, and persona main effects; H_5 is the structure × persona interaction. Studies 2 and 3 add their own pre-registered hypotheses below (Protocols 09 and 13).

Substitution hypothesis (H_1): The test-time compute framework (Snell et al., 2024) motivates the prediction that structure can substitute for model tier at the lower-tier end of the compute-optimal curve. We test this prediction with a factorial tier × structure design, priced in dollars, on open-ended tasks where no verifier exists. If the hypothesis holds, team structure should raise quality more for cheaper models than for frontier models.

Main-effect hypotheses (H_2, H_3): The registered protocol also specifies the tier main effect (H_2: frontier models score higher than their lower-tier siblings, averaged over structure and persona) and the structure main effect (H_3: teams score higher than solos, averaged over tier and persona). We report both as secondary context for the economic analysis.

Persona hypotheses (H_4, H_5): Our prior work (Hariri, 2026b) showed that cultural personas shift quality within single-model cells. H_4 extends the registered claim to this design: persona-loaded conditions should score higher than no-persona conditions, averaged over tier and structure. H_5 adds that the persona effect should be larger for team structures than for solo structures (personas amplify structural rigor). If personas primarily add stylistic variation without substance, the persona-loaded cells should instead be Pareto-dominated by their no-persona twins.

Refinement hypothesis (Study 2, Protocol 09): The self-refinement literature (Shinn et al., 2023) suggests that a low-cost critique-and-revise pass can recover quality gaps. Study 2 tests whether a frozen, add-only refiner can close the residual gap between orchestrated lower-tier outputs and frontier outputs — and whether the mechanism is additive coverage (more structure) or compressed insight (better density). Study 2

was pre-registered as a separate study before any refinement run, with its own feasibility gate (G1), superiority gate (G2), and three directional criteria (H2.1–H2.3) frozen in the decision log (Protocol 09, decisions D39–D40).

Authorship hypotheses (Study 3, Protocol 13): Every pipeline in the literature above — and in Studies 1–2 — is *human-designed*; the registered-but-unexecuted A7 arm isolated authorship as the one design variable left untested, and Study 3 executes it. Across the same five families, the cheap model either authors and runs its own workflow (D-SELF) or runs one authored by the frontier flagship (D-FRONT). The primary hypothesis is *directional*: frontier-authored pipelines outperform self-authored ones (H-D1: D-FRONT > D-SELF), on the ground that design capability transfers through authored structure — the same logic by which a cross-family refiner beat within-family editors in the Study 2 extension. Two further hypotheses complete the set: that model-authored pipelines are at least as good as the frozen human-designed instrument (H-D2, two-sided), and that frontier-authored design stays cheaper per quality point than frontier solo once design tokens are charged (H-D3, economic). A pre-registered confirmatory replication (Protocol 16) then converts the family-level effects into binding directional gates, testing whether the cost–quality ordering transports to fresh problems.

Section 3 specifies the pre-registered design, apparatus, and analysis plan that test this hypothesis set, with every run priced in dollars; Section 4 reports the outcomes. One hypothesis is mechanistic rather than purely economic: the qualitative coding protocol (Section 3.6.5) tests the refinement hypothesis's prediction that orchestration buys *coverage* while the frontier wins on *density*, by coding original, refined, and frontier answers against a frozen codebook.

3 Methodology

3.1 Research Design

The experiment is a $2 \times 2 \times 2$ between-cell factorial with model *family* as a blocking factor. The three manipulated factors are **tier** (frontier flagship vs. lower-tier sibling within the same generation), **structure** (solo single call vs. three-step team of builder → critic → refiner), and **persona** (none vs. a fixed cultural-persona mapping applied to each team role). Five model families — Kimi, Qwen, GPT-5.6, Claude, and Gemini — supply both frontier and cheaper tiers. Thirty business problems, stratified across knowledge-work domains, are identical across all cells; no instruction wrappers are added to the rendered prompt. Every design and statistical decision is time-stamped, numbered (D1–D54), and registered in a frozen decision log; twelve protocols and their amendments were registered before the corresponding computation ran, maintaining the confirmatory/exploratory separation required by preregistration discipline (Nosek et al., 2018).

Two terms recur throughout the paper and are worth fixing precisely. A **protocol** is a single pre-registered experiment plan: it states the research question, the hypotheses, the models and problems to be run, the analysis code, and the stopping rules — and it is frozen (time-stamped and registered on OSF) *before* any data for that question are generated, so that results cannot be retrofitted to expectations. An **arm** is one experimental condition inside a protocol — one specific combination of model tier, pipeline structure, and persona setting that produces a batch of runs (e.g. "lower-tier team without persona" is one arm). Appendix B lists every protocol and arm with its role and status.

This paper reports three studies and two pre-registered replications. **Study 1** is the factorial experiment above (H_1–H_5; Protocols 01–04 and Amendments A1–A4). **Study 2** asks whether a cheap *refinement pass* on an existing answer can close the remaining gap to frontier quality (Protocol 09), followed by qualitative coding of what orchestrated and frontier outputs actually contain (Protocol 10). A pre-registered **Study 2 extension** (Protocol 11) repeats the refinement test with two further families (GPT, Gemini) and adds a refiner-identity contrast: each family refined by its own model versus a *cross-family* refiner editing another family's output. **Study 3** manipulates pipeline design authorship with model-authored pipelines (Protocol 13, Section 4.3). The two replications stress-test the core claims out-of-sample: a confirmatory wave on 30 fresh problems across ten families (Protocol 12, OSF jg49x; Section 4.1.7), and a transport test of Study 3's family-level effects on new problems (Protocol 16; Section 4.3.1).

Table 2: OSF registration chain (Open-Ended Registrations branched from project node `8pwej`). The text cites Protocol numbers; this table is the authoritative GUID map.

Study / Protocol	OSF	Registered	Role
Study 1 (Protocols 01–04, A1/A2)	mjb8x	2026-08-06	factorial pre-registration[†]
Study 2 extension (Protocol 11)	83c9u	2026-08-22	refinement extension (GPT, Gemini)
Wave (Protocol 12)	jg49x	2026-08-23	Study 1 confirmatory replication
Study 3 (Protocol 13, v2)	bwc93	2026-08-29	authorship study (operative)[‡]
Study 3 (Protocol 13, G1)	639eq	2026-08-29	feasibility-gate clarification
Study 3 replication (Protocol 16)	zkm8v	2026-08-30	family-moderation transport test

[†] mjb8x *memorializes* the Study 1 pre-registration: the protocol and decision log were frozen before data collection, but this OSF timestamp post-dates main-run generation (as the registration itself discloses).
[‡] bwc93 is the operative Study 3 registration, superseding the v1 registration (9ukjz) after a pre-analysis instrument audit; the 73 v1 runs were archived and never pooled.

Registered but unexecuted arms (disclosure). The Study-1 pre-registration also specified four special mechanism arms — A5 (a critic-gated WHILE loop on the cheap team), A6 (a cost-matched frontier-solo baseline), A7 (an adaptive per-problem pipeline designer), and A8 (a shuffled persona-to-role mapping) — each with a pre-registered feasibility/futility gate. These gates were never evaluated and the arms were never executed within Study 1: the main run was scoped to the eight cells of the factorial design itself (2 tiers × 2 structures × 2 persona settings, listed in Appendix B), and the Study-2 extension and the replication wave took budgetary priority. Per no-file-drawer discipline we disclose their registered-but-unexecuted status here rather than silently dropping them. The design-authorship question behind A7 has since been executed as Study 3 (Protocol 13; Section 4.3); the loop question behind A5 is the subject of a separate pre-registered protocol (Section 8).

3.2 Models Evaluated

3.2.1 Frontier Models (Five Families)

Each family contributes its current *mainline* flagship, pinned on 2026-07-30: `kimi-k3`, `qwen3.7-max-2026-06-08`, `gpt-5.6-sol`, `claude-opus-5`, and `gemini-3.1-pro-preview`. Above-mainline tiers (e.g. `claude-fable-5`) are deliberately excluded because they would break cross-family comparability. Gemini's preview status carries a pre-registered fallback

Figure 1. Experimental Design: 2 × 2 × 2 Factorial + Apparatus

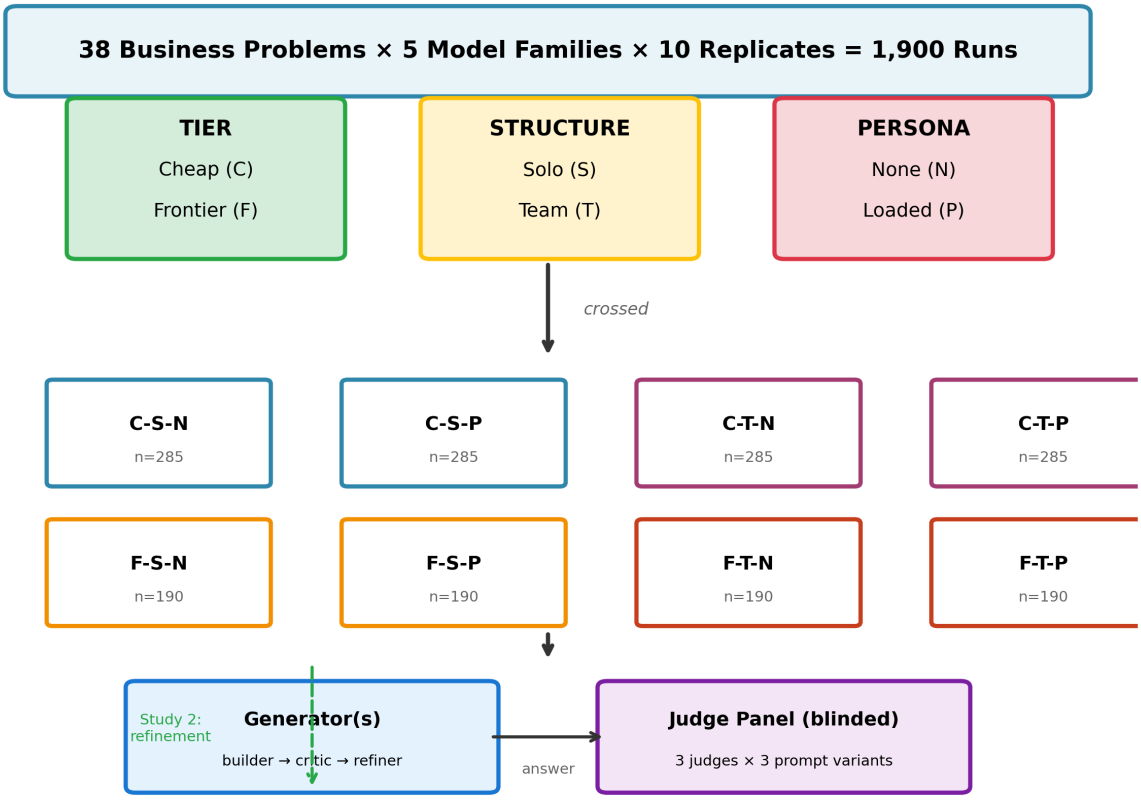


Figure 1: Experimental Design: 2 × 2 × 2 Factorial + Apparatus.

to `gemini-2.5-pro` if the model is retired mid-experiment; the fallback was never invoked. All models run with generator temperature = 1, a uniform value forced by the OpenAI and Moonshot APIs (which reject other values) and applied identically across families for *ceteris paribus* comparability (D27).

3.2.2 Lower-Tier Siblings

Each family contributes one lower-tier model from the same generation: `kimi-k2.6`, `qwen3.7-plus-2026-05-26`, `gpt-5.6-luna`, `claude-haiku-4-5-20251001`, and `gemini-3.1-flash-lite`. Price ratios (frontier : cheaper) range from 3.4× to 8× across families. The size of this price gap differs from family to family; our statistical model simply gives each family its own baseline (a *family random effect*, Section 3.6.2), so these differences in price gap cannot distort the main tier × structure comparison.

3.2.3 Replication-Wave Models (Five Additional Families)

The confirmatory wave (Protocol 12, Section 4.1.7) repeats the three core cells (C-S-N, C-T-N, F-S-N; persona excluded by design, D50) on 30 fresh problems and adds five families beyond the core set, pinned on 2026-08-22/23 per Protocol 12 §3: Thinking Machines (`Inkling` / `Inkling-Small`), OpenAI open-weight (`gpt-oss-120b` / `gpt-oss-20b`), Google open-weight (`gemma-4-31B-it` / `gemma-3n-E4B-it`), Mistral (`mistral-medium-2604` / `mistral-small-2603`), and MiniMax (`MiniMax-M3` / `MiniMax-M2.7`). The five core fam-

ilies run their pinned Study-1 models unchanged, with one disclosed routing-only deviation: the Kimi lane runs on the native Moonshot provider because the Together-hosted `kimi-k2.6` was retired before the wave (D48; identical prompts and decoding, provider route auditable in the ledger).

3.3 Orchestration Structures and Personas

Solo runs invoke a single model call on the rendered problem. **Team** runs execute a three-step pipeline — builder produces a draft, critic critiques it, refiner produces the final answer — with all intermediate outputs discarded from the judged answer (and therefore from the quality score), while the cost ledger records every step: the per-run token total is the sum of the builder, critic, and refiner calls, so team costs include the full pipeline, not just the final step. The one gap is retry accounting: failed step attempts consumed API tokens but logged zero, so on runs with a retried step (110 of 950 completed team runs) the recorded cost understates the true cost; we bound this in Section 7. Persona is implemented as a fixed mapping from role (builder/critic/refiner) to a cultural persona description, identical across all problems and families (Hariri, 2026b). The mapping is pre-registered and never varied per-problem; it is therefore a treatment level, not a confound.

3.4 Stimuli and Response Generation

Thirty domain-stratified business problems are used. Fifteen are inherited verbatim from a prior validated set (Hariri, 2025a); ten were drafted post-cutoff to control for contamination and validated as structurally identical; the remainder extend the grid to cover five domains equally. Each problem is rendered as `background_context + "\n\n" + business_problem` with no instructional wrapper — adding wrapper text would introduce an unmeasured prompt treatment. Generator temperature is fixed at 1 (Section 3.2). Every run is time-stamped and flagged for whether it executed inside the pre-registered preferred window (14:00–00:00 UTC), which captures the Qwen off-peak discount; window-free providers (Kimi, Together-hosted models, Western providers) are time-invariant and therefore unaffected. DeepSeek pricing was stable throughout the experiment (no peak-surge window was in effect during the execution period).

3.5 Evaluation Framework

3.5.1 Judge Model Selection

Three judges from families disjoint from every generator: `deepseek-chat` (resolved to `deepseek-v4-flash`), `zai-org/GLM-5.2`, and `meta-llama/Llama-3.3-70B-Instruct-Turbo`. The disjointness rule (D33) guards against self-enhancement bias (Zheng et al., 2023): even when deepseek is the registered fallback generator, the fallback was never invoked, so disjointness holds in the executed data.

3.5.2 Scoring Rubric

Five 1–5 dimensions are scored: **Novelty, Usefulness, Flexibility, Elaboration, and Cultural Sensitivity**. The composite primary outcome is the mean of the five dimensions. Three prompt variants are used per judge, frozen verbatim and never re-

optimized after a prior judge-prompt optimization attempt collapsed κ in earlier work (Hariri, 2025a). Every solution is scored by all three judges under all three variants (nine scores per solution), with candidate order shuffled per variant to neutralize position bias (Zheng et al., 2023). Judges run at temperature 0 for determinism.

3.5.3 Inter-Rater Reliability

Pairwise judge correlations on per-run composite scores are 0.87 (deepseek–zhipu), 0.61 (deepseek–meta), and 0.49 (meta–zhipu). The panel-level reliability exceeds any single judge's correlation with the latent quality; this mirrors the jury logic of Verga et al. (2024) in which diversity of smaller judges outperforms a single large judge.

3.6 Statistical Analysis

3.6.1 Effect Sizes

Welch's d with bootstrap 95% confidence intervals (5,000 resamples, seed 42) throughout (Lakens, 2013). Estimation-first reporting: intervals, not p-values, carry the weight of the argument; p-values are reported for completeness and to comply with sequential-analysis boundaries (Section 3.6.2).

3.6.2 Inferential Statistics

The primary test of H_1 is the pre-registered interaction contrast $d = (FT - FS) - (CT - CS)$, expected negative (team structure helps cheaper models more than frontier models). Inference uses a mixed model with random effects for *problem* and *family*, solving the pseudo-replication identified in the author's prior work (Hariri, 2026b). Sequential analysis with an O'Brien–Fleming spending function (protocol 03) governs interim looks and the final stopping boundary; the Bayesian JZS Bayes factor is reported as recommended and pre-registered in the robustness battery[1] but does not gate the decision. Family-wise corrections within secondary hypotheses use Benjamini–Hochberg FDR ($\alpha = 0.05$); Holm is used for the small set of directional family comparisons in Study 2.

Later studies and replications follow the same estimation-first discipline with study-specific designs. The **Study 2 extension** (Protocol 11) uses paired within-problem tests with Holm adjustment across arms, plus a two-sided refiner-identity contrast (no pre-committed direction). The **wave** (Protocol 12) fits an LMM with random intercepts for family and variance components for problem and family × problem, evaluated against binding gates W1–W7; when the dual-variance-component model was numerically singular, a pre-declared fallback substituted a parsimonious structure with bit-identical fixed effects (D52). **Study 3** (Protocol 13) tests the authorship contrast with paired t-tests within family × problem, Holm-adjusted across the five families, Novelty as the primary dimension and the composite as a robustness variant (conclusions are DV-invariant). **Protocol 16** converts Study 3's point estimates into binding directional gates: problem-level Welch tests of D-SELF against the frozen wave F-S-N baseline, one-sided p by normal approximation, and a TOST equivalence gate for GPT-5.6 (90% CI within ±0.30), executed by a hash-locked analysis script frozen before judging.

[1] Amendment A2 — Extended Interim-2 Analysis Methods, items A2.1–A2.7; OSF registration https://osf.io/mjb8x.

3.6.3 Economic Analysis

Cost is stored as tokens in a run-level ledger and re-priced through a frozen price vector (`config/price_vector.json`, version 2026-07-30). Because the ledger does not split input from output tokens, cost is computed under a blended price $\text{blend}(\varphi) = \varphi \cdot p_{\text{out}} + (1 - \varphi) \cdot p_{\text{in}}$ with a primary $\varphi = 0.50$ and a sensitivity panel $\varphi \in \{0.25, 0.50, 0.75\}$ (Diewert 1976 provides the price–quantity decomposition precedent; Steegen et al. 2016 the multiverse-robustness precedent). Frontier-membership claims must be stable across all three φ. The **Tiered Efficiency Gain Ratio (T-EGR)** is $(\text{Quality}_{\text{cell}}/\text{Quality}_{F\text{-}S\text{-}N})/(\text{Cost}_{\text{cell}}/\text{Cost}_{F\text{-}S\text{-}N})$, computed within each family and pooled; F-S-N is the baseline. **ICER** is reported relative to the lowest-cost cell. Judge cost is excluded from unit T-EGR (it is shared infrastructure, not cell compute; a disclosed exclusion).

3.6.4 Robustness Checks

The pre-registered battery (Amendment A2, items A2.1–A2.7) comprises bootstrap confidence intervals on all primary contrasts (A2.1), Benjamini–Hochberg FDR on secondary contrast families (A2.2), a Type-II factorial ANOVA sensitivity layer with partial η^2 (A2.3), JZS Bayes factors for non-significant contrasts (A2.4), leave-one-family-out sensitivity (A2.5), leave-problems-out generalizability (A2.6), and a sensitivity power analysis in place of post-hoc power (A2.7). After the data lock, an explicitly exploratory amendment (A3, registered in the decision log but not part of the OSF package) added two analyses: empirical-Bayes partial pooling of per-family interaction estimates (A3.3) and a random-effects meta-analysis across the five families (A3.4). All A3 results are labeled exploratory wherever reported. The meta-analysis confirms the interaction generalizes (pooled $d = -0.270$, $I^2 = 28.5\%$).

3.6.5 Qualitative Coding Protocol (Study 2)

Thirty triples (original CTP / refined CTP / same-problem same-family FSN) are sampled deterministically (seed 20260801, 10 per family). Two blinded LLM coders perform a two-wave coding: free induction on a subset, then closed coding against a frozen 13-code codebook.[2] The codebook is organized around three research questions: **RQ1** asks whether orchestration adds *insight* (compressed reasoning, named concepts, diagnostic reframing) — six codes; **RQ2** asks whether orchestration adds *structure* (scaffolding, timelines, matrices) — six codes; **RQ3** asks whether orchestration adds *bloat* (redundancy) — one code. Cohen's κ gate is 0.70; codes below threshold are downgraded to medium reliability and disclosed (Cohen, 1960). The prompt presented to coders contains the three texts only — no cell label, family, or score, preserving blind evaluation.

3.7 Validity and Reliability

Construct validity: the rubric separates Novelty from Usefulness/Elaboration, so a stylistic shift (e.g. more discourse markers from a persona) is not conflated with substantive quality. **Internal validity:** frozen configurations, blinded judging, shuffled order, disjoint judge families. **External validity:** single task domain (English business strategy) is the acknowledged boundary. **Disclosed threats.** (i) *Uncounted intermediate*

[2] Codebook development followed the pre-registered qualitative analysis protocol (OSF registration https://osf.io/mjb8x, file `02-QUALITATIVE-ANALYSIS-PROTOCOL.md`).

outputs. Team pipelines generate intermediate outputs (drafts, critic feedback) on the way to the final answer; these are not included in the cost ledger, so reported team costs are slightly understated. The bias favors the team cells — all lower-tier configurations — and therefore favors the paper's central claim. It is disclosed but not corrected, since correcting it could only shrink, not reverse, the cost advantage. (ii) *Truncation at source.* A small set of very long runs was capped by the transport layer before storage; each run records whether this happened, and the affected runs are audited separately rather than silently excluded.

3.8 Code and Data Availability

All analysis code, the frozen token ledger, judge rationales, the decision log (D1–D54), and Protocols 09–10 and 13–16 are committed to the `orchestration-economics` repository. The pre-registration trail is the OSF registration chain mapped in Table 2: each OSF registration archives its protocol file, and the table discloses the two exceptions — the Study 1 memorialization (mjb8x) and Study 2 (Protocols 09–10), which are committed to the repository and frozen in the decision log (D39) but were not separately OSF-registered. A materialized results layer (D54) ships every statistical test as a flat CSV — per-solution judge scores for all studies, the wave gate statistics, the heterogeneity batteries, and the qualitative codebook with per-code κ — so that no number in this paper requires re-running code to inspect. A reproduction runbook (Sandve et al., 2013) lists the commands that regenerate every number reported in Section 4.

3.9 AI-Assisted Research Disclosure

This study used large language models as research tools under continuous human supervision. Specifically, **Kimi** (Moonshot AI, `kimi-2.6 and kimi-k3`) and **Qwen** (Alibaba Cloud, `qwen3.7-max and qwen3.8-max`) were employed in the following capacities: (1) **ideation** — brainstorming experimental design alternatives, hypothesis formulations, and analysis strategies during the protocol-drafting phase; (2) **code generation** — producing initial versions of the execution pipeline, statistical analysis scripts, and figure-generation code, all of which were reviewed, corrected, and approved by the author before execution; (3) **audit** — reviewing citation accuracy, checking statistical assumptions, and identifying inconsistencies across protocol documents; (4) **execution** — running the generation and judging campaigns through API calls, with all configurations frozen and logged before each run. Higher-tier models were employed for ideation, audits, and task execution; lower-tier models were employed for code generation.

No LLM made unsupervised decisions about the experimental design, statistical analysis, or the interpretation of results. All twelve pre-registered protocols, all 54 design decisions (D1–D54), and all analysis choices were authored, reviewed, and approved by the human author. Outputs were validated against expected ranges, and every reported number traces to a frozen, version-controlled analysis pipeline. The qualitative coding protocol (Section 3.6.5) used two LLM coders with blinded evaluation — a design choice that converts the LLM's systematic biases into a measured inter-rater reliability coefficient ($\kappa = 0.824$) rather than an unmeasured confound.

A concrete oversight episode illustrates why continuous human audit is not procedural decoration. When the AI agent implemented the Study 3 design-authorship instrument, it silently introduced two deviations from the product feature under study (DeeWaanAI's

auto-designer): it hard-capped pipelines at four steps (the platform allows up to seven) and wrote a bespoke designer prompt, whereas the platform uses a fixed internal prompt with free-form, task-adapted roles. Both deviations were invisible in the code's behaviour — every pilot design complied, so no error surfaced — and were caught only when the human author audited the instrument's rationale against the platform source code before confirmatory data collection. The instrument was corrected and the amendment pre-registered *before* the Study 3 campaign (D56). The lesson for researchers delegating to AI agents: an AI assistant's silent defaults are themselves experimental-design decisions, and they diverge from the intended construct precisely where the researcher did not specify — audit the rationale behind every parameter, not only the parameters named in the protocol.

The use of LLMs as evaluation judges (Section 3.5) is a distinct methodological choice from the use of LLMs as research tools: the judges evaluate *generated business solutions*, not the research process itself. The disjointness rule (D33) ensures that no judge shares architecture with any generator, mitigating the self-enhancement bias documented by Zheng et al. (2023).

Platform use and conflict-of-interest disclosure. All pipeline generation and execution in Studies 1–3 ran on **DeeWaanAI** (`https://deewaanai.com`), a commercial orchestration platform; the auto-designer evaluated in Study 3 and Protocol 16 is DeeWaanAI's production feature, used unmodified (fixed internal design prompt, free-form roles, up to seven steps with critic-gated loops). The author is affiliated with DeeWaanAI, and the findings directly inform the platform's routing and persona features (Section 9). This relationship is disclosed so readers can weight the translational claims accordingly; all protocols, gate outcomes (including failed ones), and attrition events are pre-registered and reported as-is, and the full data and code are publicly available.

4 Results

This section reports three studies that escalate a single business question — *orchestrate cheaper models, or pay for the frontier?* **Study 1** (Section 4.1) tests whether pipeline structure substitutes for model tier and prices the answer in dollars. **Study 2** (Section 4.2) explains *why* that substitution stops short of the frontier. **Study 3** (Section 4.3) asks whether it matters *who designs* the pipeline. Each study's headline claims are then stress-tested out-of-sample in a pre-registered replication, so the rules we recommend are the ones that survived contact with new data — and the ones that did not are reported as failures, not buried.

4.1 Study 1: Does Structure Substitute for Model Tier?

4.1.1 The interaction effect: structure is a lower-tier amplifier

From a resource-allocation perspective, the most important finding is that pipeline structure delivers higher marginal returns at the lower tier than at the frontier tier. The pre-registered interaction contrast is $d = -0.227$ (bootstrap 95% CI $[-0.410, -0.040]$, Welch $t = -2.45$, $p = 0.0145$), with the sequential-analysis boundary crossed at the scheduled final look. For a firm making an AI budget decision, this implies that pipeline dollars are better spent upgrading lower-tier models to team configurations than adding team

overhead to already-expensive frontier models. The JZS Bayes factor for the interaction is 1.002 (inconclusive), reported as recommended but not gating the decision.

The secondary effects provide additional strategic context. The **tier main effect** (H_2: $d = +0.294$, CI $[+0.201, +0.382]$, $BF_{10} = 1.7 \times 10^7$) confirms that frontier models command a quality premium, but the magnitude is dwarfed by the cost difference. The **structure main effect** (H_3: $d = +0.117$, CI $[+0.030, +0.205]$) shows that teams slightly outperform solos on average — a small but consistent productivity gain from orchestration. The **persona main effect** (H_4) is large and in the *opposite* direction to the pre-registered expectation: $d = -0.691$ (CI $[-0.761, -0.620]$, $BF_{10} = 1.0 \times 10^{50}$). Persona-loaded cells score *lower* than their no-persona twins, representing a failed investment in stylistic differentiation. From a business perspective, personas degrade output quality sharply while reducing cost only marginally (persona cells emit fewer tokens) — a value-destroying configuration. An exploratory dimension decomposition (Amendment A3, item A3.5) shows that Novelty is the sole dimension where personas help ($+0.154$, $p_{\text{FDR}} = 0.001$) while Usefulness (-1.024) and Elaboration (-0.966) drive the reversal. The structure × persona interaction (H_5) is not significant ($F = 1.23$, $p = 0.27$).

> **Strategic Implication:** Pipeline structure is a lower-tier amplifier, not a universal substitute: the marginal return to team structure is higher at the lower tier, although the highest absolute return remains the lower-tier solo call (Section 4.1.4). Personas, by contrast, represent value destruction — they degrade quality while reducing cost only marginally.

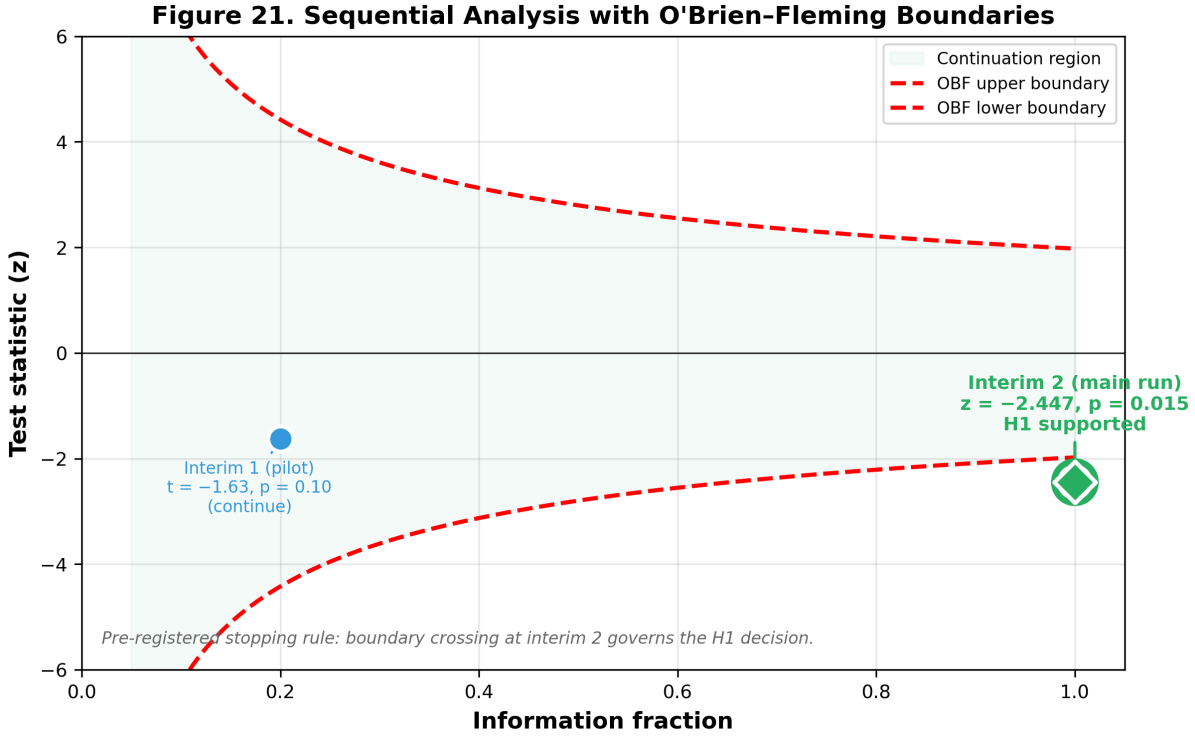


Figure 2: Sequential analysis with O'Brien–Fleming spending function. The test statistic $z = -2.447$ crosses the interim-2 boundary $|z| \geq 1.977$ (dashed line). The shaded region is the continuation region. The pre-registered decision is to stop and report H_1 as supported.

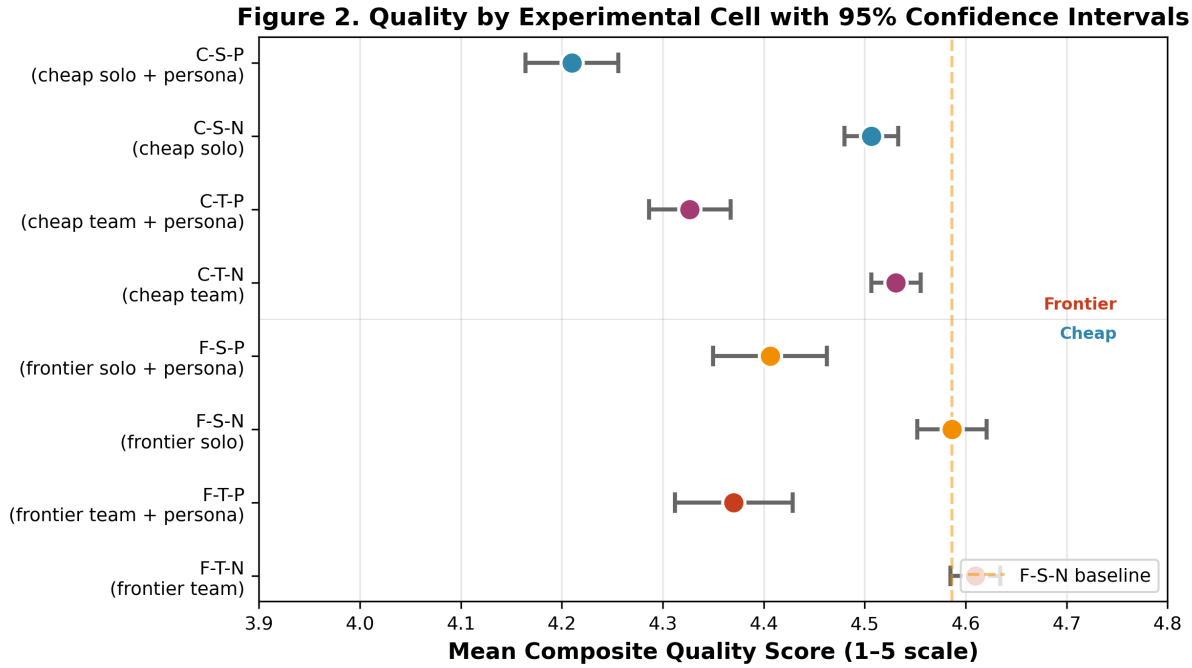


Figure 3: Quality by Experimental Cell with 95% Confidence Intervals.

4.1.2 The persona penalty: style without substance

Every persona-loaded cell except the budget-extreme cheap solo is Pareto-dominated on the cost–quality plane (Section 4.1.4): C-T-P, F-S-P, and F-T-P are each beaten on quality by lower-cost no-persona cells. Personas do *not* raise cost — persona cells emit fewer tokens and price slightly *below* their no-persona twins — but the small saving buys a large quality loss, so the twin comparison on quality is uniformly negative ($d = -0.691$). The lone survivor, C-S-P, remains on the frontier only because it is the cheapest configuration tested; its quality is the lowest of all eight cells. Its headline efficiency ratio (T-EGR 8.60, the highest in the study) is purely a small-denominator artifact of that minimal cost — it marks the budget floor, not a recommended configuration, since the same tier without a persona (C-S-N) buys 0.30 points of quality for an additional \$0.005. The penalty is largest on dimensions that rubric readers associate with substance (Usefulness, Elaboration), smallest on the one most susceptible to stylistic signaling (Novelty). This pattern mirrors the contingency finding in the author's prior work (Hariri, 2026b) and is consistent with the jagged-frontier mechanism (Dell'Acqua et al., 2023): personas push the generator toward a more stylized region of its output space that the judge, correctly, penalizes on substance.

> **Strategic Implication:** Personas degrade output quality sharply while reducing cost only marginally — they are value-destroying for knowledge-work tasks. Firms should treat them as a failed investment rather than a quality lever.

4.1.3 The frontier gap: lower-tier teams do not beat frontier solo

Paired comparisons between cheaper-team-with-persona (CTP) and the frontier-solo-no-persona baseline (F-S-N) are negative in all five families (kimi $d = -0.71$, claude $d =$

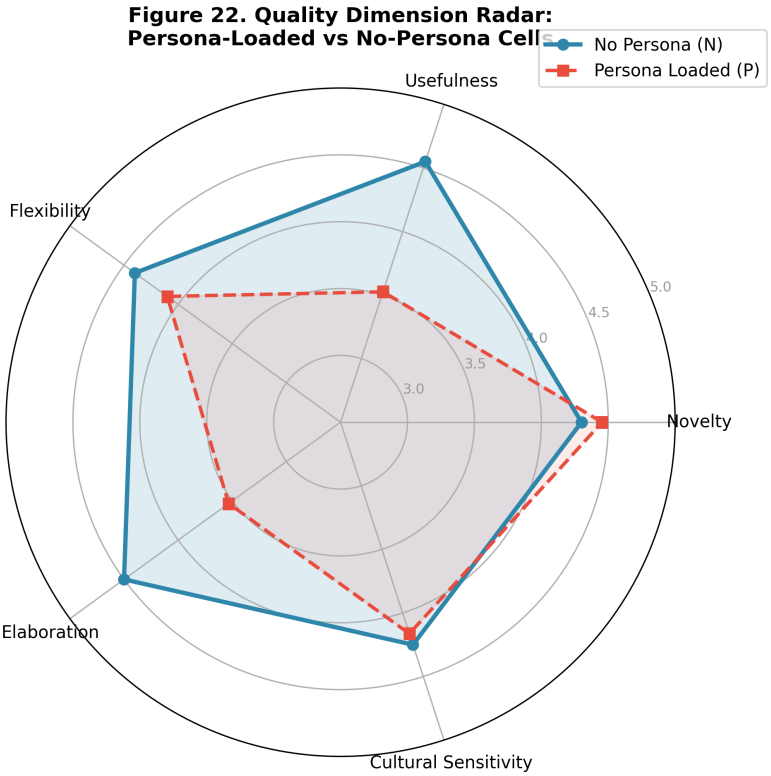


Figure 4: Quality-dimension radar for persona-loaded (red, dashed) versus no-persona (blue, solid) cells. The persona polygon is visibly smaller on Usefulness and Elaboration, slightly larger on Novelty — the visual signature of the style-over-substance tradeoff.

-1.42, qwen $d = -2.17$; all CIs exclude zero after Holm correction). This residual gap is the empirical target that Study 2 (Section 4.2) asks a cheap refinement pass to close.

> **Strategic Implication:** As executed, lower-tier teams do not beat a single frontier call on matched problems. The substitution thesis is bounded: structure improves lower-tier models but cannot fully close the frontier gap without additional investment.

4.1.4 The Pareto frontier: optimal configurations for budget-constrained firms

The cost–quality map is shown in Fig. 6. At the primary $\varphi = 0.50$, the cell means are:

The Pareto frontier is **identical across all three** φ: {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N}. The dominated set {C-T-P, F-S-P, F-T-P} contains every persona-loaded team cell — the economic echo of the Section 4.1.2 persona penalty. Frontier membership is stable across the sensitivity panel, satisfying the pre-registered claim rule.

ICER (vs. the lowest-cost cell, C-S-P): C-S-N is the best buy at +$0.016 per point of composite quality; F-T-P is the worst at +$1.81 per point. The refined kimi C-T-P point (not in the table) sits at $0.0654 / quality 4.585 — absolute parity with pooled F-S-N (4.586) at 78% of its cost, with the caveat that the paired within-family comparison remains negative (disclosure per Section 4.2.1).



Figure 5: Persona Effect by Quality Dimension (FDR-Corrected).

Table 3: Cell summary statistics (primary analysis, $\varphi = 0.50$).

Cell	n	Cost (USD)	Quality	T-EGR
C-S-P	285	0.0145	4.210	8.60
C-S-N	285	0.0194	4.507	6.15
C-T-P	285	0.0715	4.327	1.58
C-T-N	285	0.0734	4.531	1.49
F-S-P	190	0.0670	4.406	1.36
F-S-N	190	0.0835	4.586	1.00 (base)
F-T-N	190	0.2951	4.610	0.27
F-T-P	190	0.3044	4.370	0.27

Strategic Implication: The economically optimal strategy is tier substitution at the lower-tier end: a single call from a cheaper model (C-S-N) delivers 98% of frontier quality at 23% of the cost, and the three-step team variant (C-T-N) buys coverage at 88% of the cost. The worst dollar in the experiment is spent on frontier teams — they cost 3.5× the frontier-solo price for negligible quality gain.

4.1.5 Robustness across model families and problems

The pre-registered robustness battery (Amendment A2) supports the Section 4.1.1 decision. Leave-one-family-out (A2.5): d ranges from -0.143 (drop kimi) to -0.303 (drop gemini), sign stable in all five folds; no single family drives the interaction. Leave-one-problem-out (A2.6): d ranges from -0.212 to -0.247 across all 19 folds; no problem drives the interaction. Sensitivity power at the smallest cell ($n = 190$) detects $d = 0.288$ at 80% power; the observed $|d| = 0.227$ is detected through the within-subjects contrast over

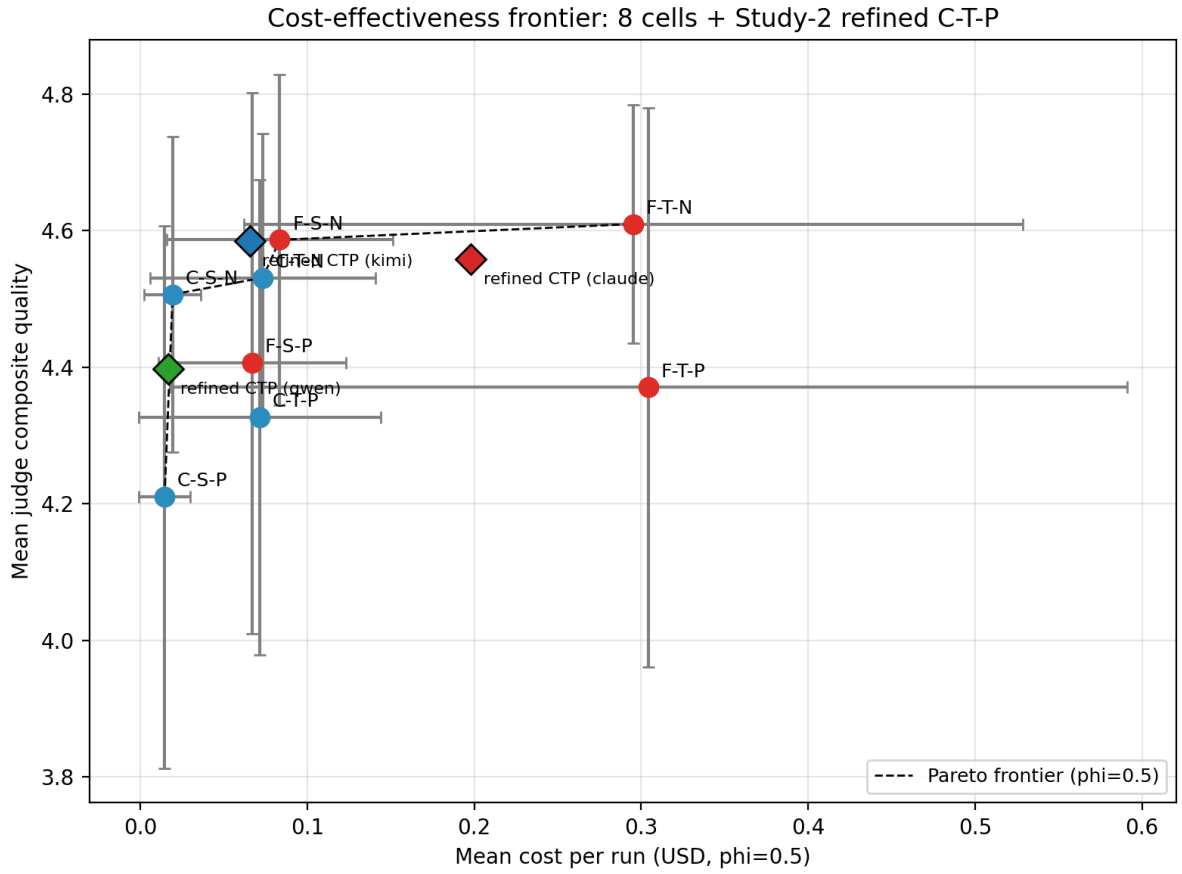


Figure 6: Cost-Effectiveness Frontier: 8 Cells + Study-2 Refined CTP.

all 1,900 runs rather than per-cell power (A2.7). The Type-II ANOVA sensitivity layer (A2.3) agrees: tier × structure interaction $F = 6.90$, $p = 0.0087$, partial $\eta^2 = 0.0036$; residuals are non-normal, so the bootstrap intervals and Welch degrees of freedom remain the authoritative inference, as pre-specified. Length adjustment on the H_1 contrast yields $p = 0.63$ — the interaction is not a length confound. The random-effects meta-analysis over the five families (exploratory, Amendment A3, item A3.4) gives a pooled $d = -0.270$ (CI $[-0.472, -0.069]$, $z = -2.64$, $p = 0.008$) with low-moderate heterogeneity ($\tau^2 = 0.015$, $I^2 = 28.5\%$), confirming the interaction generalizes.

Across all battery perturbations the interaction keeps its sign and the decision is unchanged. The leave-one-out sensitivity (Fig. 9) and random-effects meta-analysis (Fig. 10) confirm that no single family, problem, or outlier drives the result.

> **Strategic Implication:** The findings are robust to judge, model family, problem domain, and output-length variations. The interaction is small but reliable and generalizes across the five leading LLM families — a stable result for firms making platform-agnostic investment decisions.

4.1.6 Exploratory heterogeneity: domain contingency, a capability-gap mechanism, and the persona decomposition

Status declaration. Everything in this subsection is *exploratory and post-hoc*: it was formulated after data lock, is not part of the pre-registered hypothesis set H_1–H_6, and

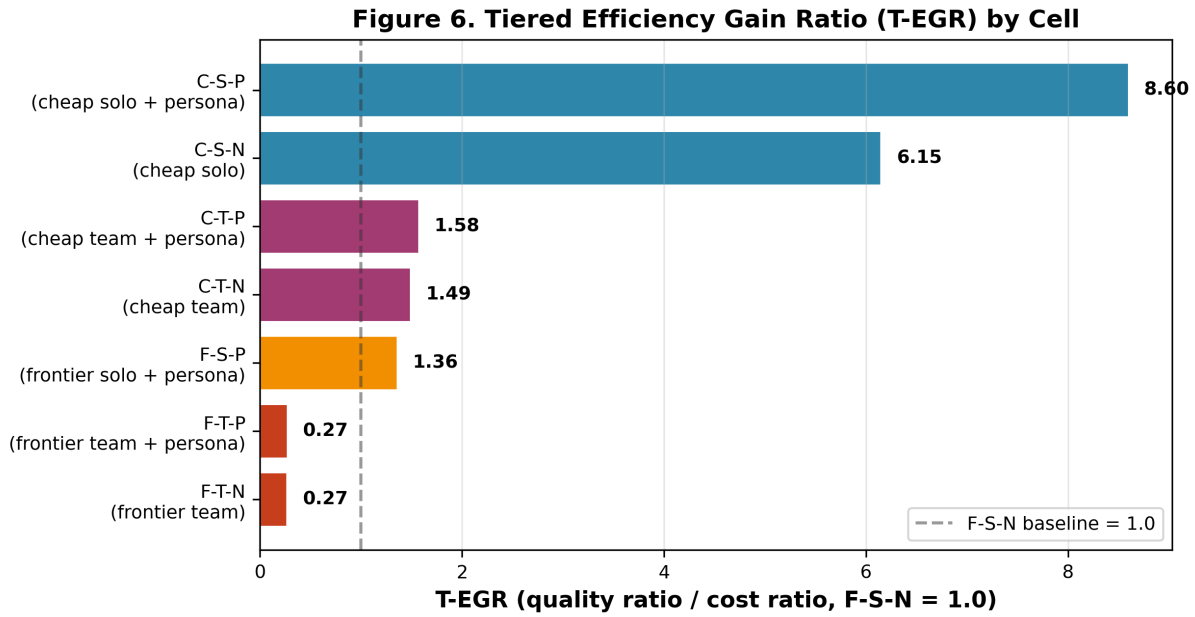


Figure 7: Tiered Efficiency Gain Ratio (T-EGR) by Cell.

consists exclusively of re-analyses of the frozen Study-1 artifacts (no new generation or judging calls). We report it because it generates concrete, falsifiable predictions — which were then tested by the pre-registered confirmatory replication wave (Section 4.1.7). Anticipating that section: **all three upgrade candidates below (domain contingency, the capability-gap mechanism, the marketing variance advantage) FAILED their binding out-of-sample gates** and therefore remain exploratory; the numbers in this subsection stand as the Study-1 record, and the wave numbers are reported alongside in Section 4.1.7 (Nosek et al., 2018; Open Science Collaboration, 2015).

Where does heterogeneity live? Three screening analyses localize the variation. Problem difficulty does not moderate the team effect ($r = 0.153$, $p = 0.53$ across the 19 problems). Family quality rankings are remarkably stable across problem domains (Kendall's $W = 0.922$; Kendall & Babington Smith, 1939), so family choice is largely domain-invariant. But measurement reliability itself is domain-heterogeneous: single-judge agreement ICC(3,1) (Shrout & Fleiss, 1979) ranges from 0.235 (HR/organization) to 0.523 (marketing), with panel reliability ICC(3,9) between 0.73 and 0.91 — HR answers are simply harder for any judge, human or LLM, to score consistently.

The team effect is domain-contingent. Pooling the cheap tier, the team-minus-solo quality effect varies from +0.004 (marketing — the only domain where teams match solo) through strategy (−0.036), HR (−0.053), operations (−0.058), product innovation (−0.103), to finance (−0.106). In marketing the team advantage extends beyond the mean: team output variance is significantly *lower* than solo (Brown–Forsythe test, $p = 0.020$; Brown & Forsythe, 1974), and downside risk shrinks across cells ($P(\text{composite} < 4.0)$: solo 2.8%, team 1.8%, frontier 1.1%). The ensemble-variance logic is classical — combining models reduces error when members are accurate and diverse (Hansen & Salamon, 1990; Dietterich, 2000) — but here it appears only for communication-intensive tasks.

A capability-gap mechanism. The family-level team effect correlates almost perfectly with the family's capability gap to the frontier (Pearson $r = -0.978$, $p = 0.004$; Spearman $\rho = -1.000$; $n = 5$ families): teams help most where the cheap model is far-

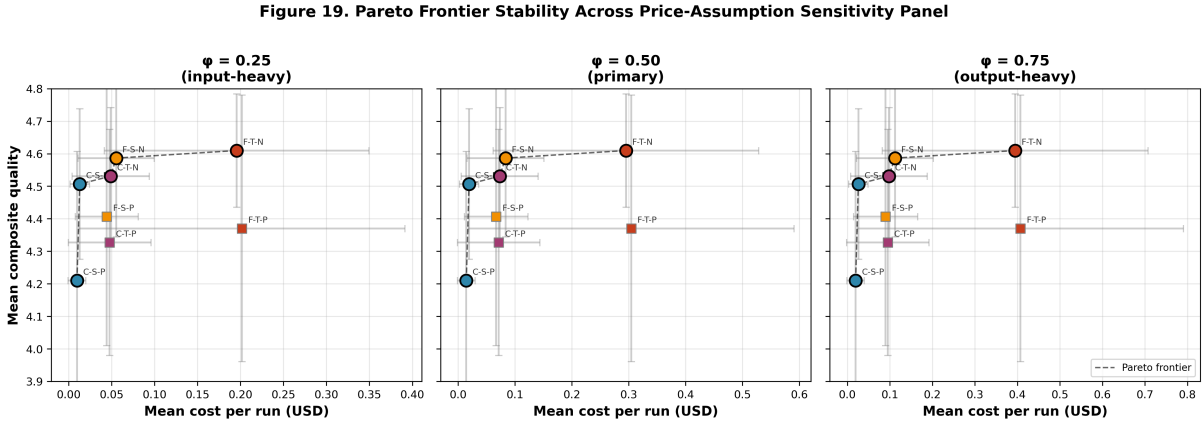


Figure 8: Pareto frontier stability across the price-assumption sensitivity panel. Left: $\varphi = 0.25$ (input-heavy blend). Center: $\varphi = 0.50$ (primary analysis). Right: $\varphi = 0.75$ (output-heavy blend). The frontier set {C-S-P, C-S-N, C-T-N, F-S-N, F-T-N} is identical in all three panels, satisfying the pre-registered cross-φ stability rule.

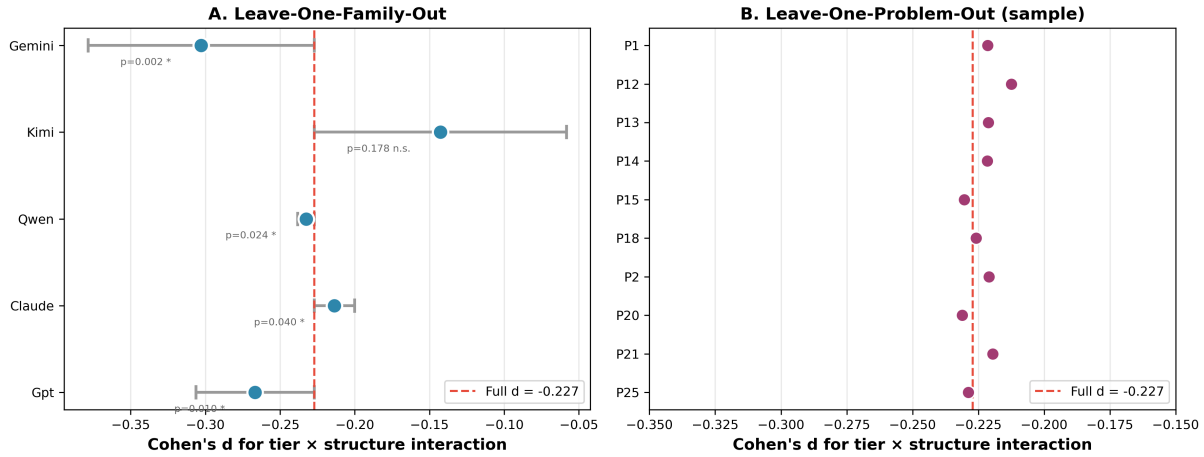


Figure 9: Robustness: Leave-One-Out Sensitivity (A2 Battery).

thest from its frontier sibling (kimi, −0.164 gap, +0.105 team effect) and hurt where the gap is small (claude, −0.357 gap, −0.292 team effect). With only five families this is a hypothesis generator, not a test; the replication wave expands the family universe specifically to power this correlation properly (moderator analysis logic as in Borenstein et al., 2009).

Length control reveals an asymmetry. A length-adjusted mixed model (composite ∼ log-tokens + cell, random family and problem intercepts; Westfall et al., 2014) leaves the frontier advantage intact (+0.078, $p < 0.001$) but reduces the team's small raw edge (+0.026) to −0.018 ($p = 0.40$). We deliberately do *not* read this as mediation: quality is judged on visible full text, and team structure mechanically produces longer visible answers, so controlling length partially controls the intervention itself. The defensible reading is the asymmetry — frontier superiority survives length adjustment; the team edge does not. The same model locates the remaining economics: frontier solo is *faster* than cheap teams (94.8s vs. 131.7s) and cheap solo dominates efficiency (295.8 vs. 28.1 quality points per dollar-minute for teams).

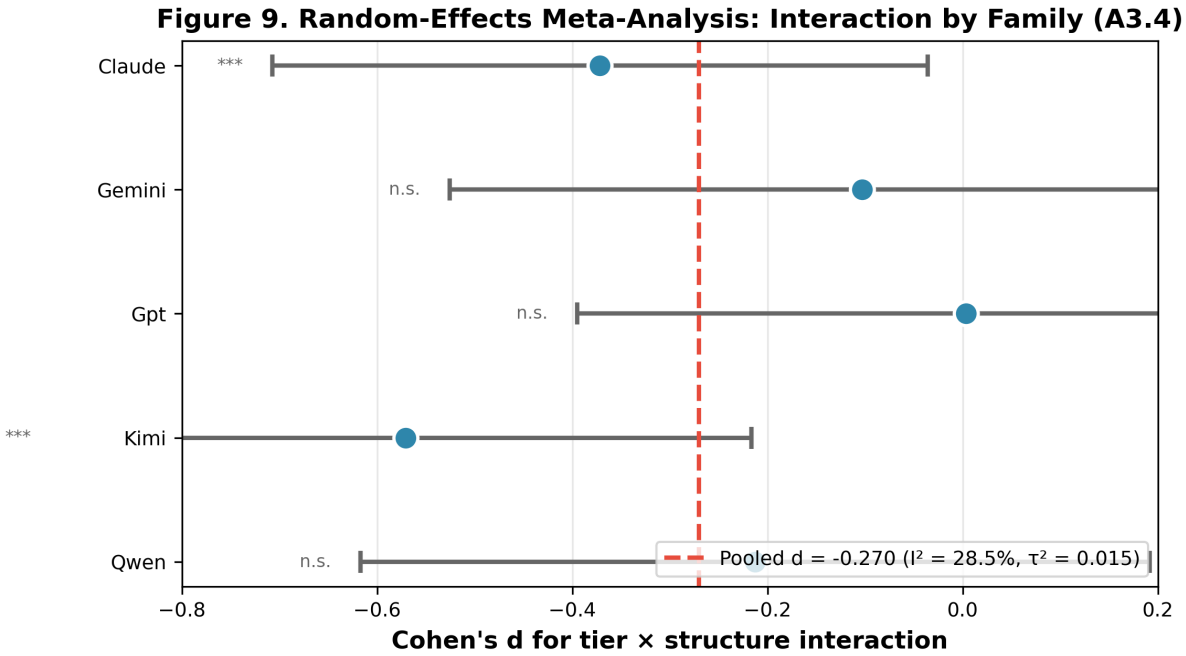


Figure 10: Random-Effects Meta-Analysis: Interaction by Family.

Persona decomposition. Extending the battery to the persona axis (full $2 \times 2 \times 2$ coverage): (i) personas produce no spillover into the judged Cultural Sensitivity dimension (pooled $d = -0.10$ to -0.05, all domain cells n.s.; the personas were designed as a general cultural-perspective injection, not as an instrument for this specific rubric item); (ii) the solo-persona novelty gain is genuine — it survives length control ($+0.465$, $p < 0.001$) and is uniform across all six domains ($+0.36$ to $+0.57$) against a uniform usefulness cost (-0.96 to -1.10); (iii) teams destroy that gain (structure × persona interactions: Novelty -0.693, Usefulness $+0.711$, both $p < 0.001$, Holm-corrected; Holm, 1979); (iv) persona answers are judged *more* consistently (panel ICC(3,9) 0.933 vs. 0.918), so the effect is not a measurement artifact; and (v) personas roughly double outcome variance (solo SD $0.232 \rightarrow 0.397$), consistent with a style-injection account. Length bias in LLM judging is a documented threat (Zheng et al., 2023; Dubois et al., 2024), which is why each effect above was re-tested under length control before reporting.

> **Strategic Implication (exploratory — gate FAILED):** The contingency picture above did *not* survive the replication wave (Section 4.1.7): on fresh problems the frontier advantage is uniform across domains, the capability-gap correlation vanishes ($r = -0.21$ vs. -0.978), and the marketing variance advantage reverses. We retain it as a hypothesis generator only; the decision framework of Section 5.1 rests on the replicated core claims, not on these.

4.1.7 Out-of-sample confirmation: the replication wave (Protocol 12)

Design. Protocol 12 (pre-registered, OSF jg49x, before any wave generation) froze a binding upgrade rule: a Study-1 *exploratory* finding becomes a headline claim only if its pre-registered wave test passes at the pre-registered alpha and direction; the core claims are re-tested with the same frozen instruments. The wave uses 30 fresh post-cutoff business problems (a second 6×5 domain grid, zero overlap with Study 1), the three core

cells (C-S-N, C-T-N, F-S-N), and a family universe doubled from 5 to 10 (adding Mistral, MiniMax, Gemma, GPT-OSS, and ThinkingMachines): 2,400 generation runs, judged by the same frozen three-judge panel under the same three verbatim prompt variants (21,762 judgment cells; 2,418 analyzed runs). The analysis script was frozen and hash-locked before judging; a disclosed numerical fallback (a singular dual variance-component LMM, replaced under a pre-declared rule that hard-requires bit-identical fixed effects) is recorded in the results artifacts and our decision log (D52). Provider events are disclosed, not imputed: the Gemini lane was deferred on a free-tier daily cap until a billing upgrade, and the Kimi frontier lane was completed via a Together.ai routing deviation (same pinned model version; D53) after Moonshot credit exhaustion.

Gate outcomes. Table 4 reports the complete-panel results (the interim 9-family record of 2026-08-24 stands unchanged in the wave report; the complete panel adds Gemini and the Kimi remainder).

Table 4: Confirmatory replication wave: gate outcomes on the complete 10-family panel ($n = 2{,}418$ judged runs, $\alpha = 0.05$).

Gate	**Result (complete panel)**	**Verdict**
W1 team effect (C-S-N → C-T-N)	$+0.0168$, $p = 0.002$	PASS
W2 upgrade threshold (C-T-N → F-S-N)	$+0.0745$, $p < 0.0001$	PASS
W3 domain contingency	interaction LRT $p = 1.0$; directional signs 5/6; *no* positive domain (marketing flips negative)	FAIL
W4 capability-gap mechanism	$r = -0.214$, $p = 0.55$ ($n = 10$ families; power for $\rho = 0.80$ is 0.83) vs. Study-1 $r = -0.978$	FAIL
W5 marketing variance advantage	Brown–Forsythe $p = 0.795$, direction reversed ($\mathrm{SD}_{\mathrm{team}}$ 0.272 > $\mathrm{SD}_{\mathrm{solo}}$ 0.263)	FAIL
W6 economics	F-S-N 40.6 > C-T-N 21.2 quality/$-min; Pareto = {C-S-N, F-S-N}; T-EGR 4.04 (C-S-N), 0.96 (C-T-N)	PASS
W7 judge reliability (audit)	ICC(3,1) 0.471–0.622 by domain	—

Reading. The three core claims replicate out-of-sample on the full family universe: (i) cheap teams beat cheap solo by a small but real margin (W1); (ii) frontier solo beats cheap teams decisively on quality (W2, an effect $\sim 4.4\times$ the team effect); (iii) the cost–quality frontier is bimodal — cheap solo for volume economics (T-EGR = 4.04), frontier solo for quality — with cheap teams the dominated middle at current list prices (W6; the ordering is unchanged under the MiniMax promotional-price sensitivity vector, T-EGR 4.11/0.98). All three exploratory upgrade candidates fail: the "marketing exception" was a sample-specific fluctuation (marketing flips negative, $\Delta = -0.075$), the near-perfect capability-gap correlation was an $n = 5$ artifact, and teams do not reduce downside variance in any domain. We read the failed gates as the design working as intended: the pre-registration separated what survives contact with new data from what does not (Open Science Collaboration, 2015; Camerer et al., 2018), and no post-hoc relabeling is permitted in either direction.

Key Takeaway: the headline economics of this paper are replication-grade; the contingency nuances are not. Practitioner rules should be built on W1/W2/W6 only.

4.2 Study 2: Why Substitution Stops Short of the Frontier

Study 1 established that orchestrated lower-tier models approach but do not reach the frontier (Section 4.1.3). Study 2 asks the follow-up a budget owner actually faces: *can a cheap refinement pass close that residual gap — and if it cannot, why not?* The answer is what separates a risk-mitigation strategy from a replacement strategy.

4.2.1 The refinement ceiling: closing the gap at low marginal cost

A frozen, add-only refiner is applied to the CTP outputs. The pilot satisfied the feasibility gate (G1, 100% completion); the locked pool comprised all 171 matched CTP–FSN pairs (57 per family), of which the pilot covered 20 per family and a pre-registered extension (registered before execution) covered the remainder. The pre-registered superiority gate (G2: $d \geq 0.2$ with CI excluding 0 in all three families, in the direction refined > FSN) is mechanically failed in all three families: kimi $d = -0.295$, claude $d = -1.301$, qwen $d = -0.809$ (all CIs strictly negative after Holm). The superiority claim is formally rejected.

Refinement nevertheless does substantial repair work. Measured as gap recovery against each family's own pre-refinement CTP-vs-FSN gap, the pass recovers 60% of the gap for kimi, 8% for claude (whose pre-refinement gap was already small), and 64% for qwen. The Novelty protection criterion (Protocol 09 criterion H2.2: no family loses more than −0.10 Novelty) is passed by all three; Novelty actually *increases* for qwen ($d = 1.10$, $p_{\text{FDR}} < 0.001$). Usefulness and Elaboration show large gains in kimi and qwen ($d = 0.57$ to 1.35, all $p_{\text{FDR}} < 0.001$) and are unchanged in claude — which was already at ceiling on those dimensions before refinement.

Length analysis confirms the dilution hypothesis is *not* the driver: the pooled Pearson correlation between length ratio (refined/original) and composite-score change is $r = +0.68$, Spearman $\rho = +0.71$ ($p < 0.001$). Longer refined answers score higher, not lower.

Cross-family comparisons (Fig. 12) remain negative after refinement: no cross-family CTP pairing flips to superiority, confirming that the within-family result is not an artifact of family selection.

Strategic Implication: A $0.002 refinement pass delivers bounded improvement — it recovers most of the frontier gap on average but never achieves paired superiority. For firms, this means refinement is a viable risk-mitigation strategy (improve quality at lower cost) but not a replacement strategy (it will not beat the frontier).

4.2.2 Why orchestration buys coverage but not density

Thirty triples (10 per family) are coded under a 13-code codebook. Pooled Cohen's $\kappa = 0.824$ (93.1% agreement), well above the 0.70 gate. Six of the thirteen codes fall below 0.70 (`diagnostic_reframe` 0.54, `named_concepts` 0.49, `assumption_test` 0.56, `structured_report` 0.59, `external_authority` 0.56, `redundancy_bloat` 0.65), a base-rate artifact: percentage agreement on these codes ranges from 82% to 98%, but κ is

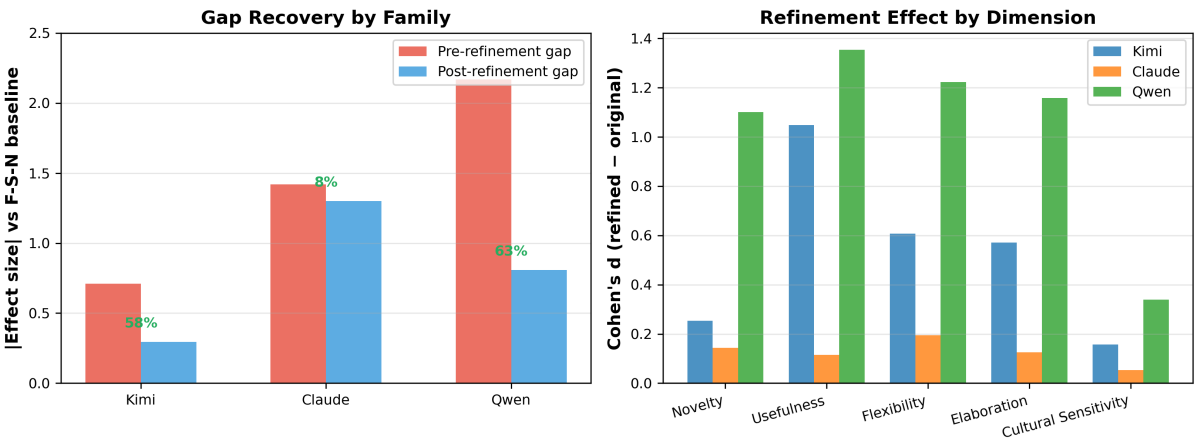


Figure 11: Study 2: Refinement Probe Results.

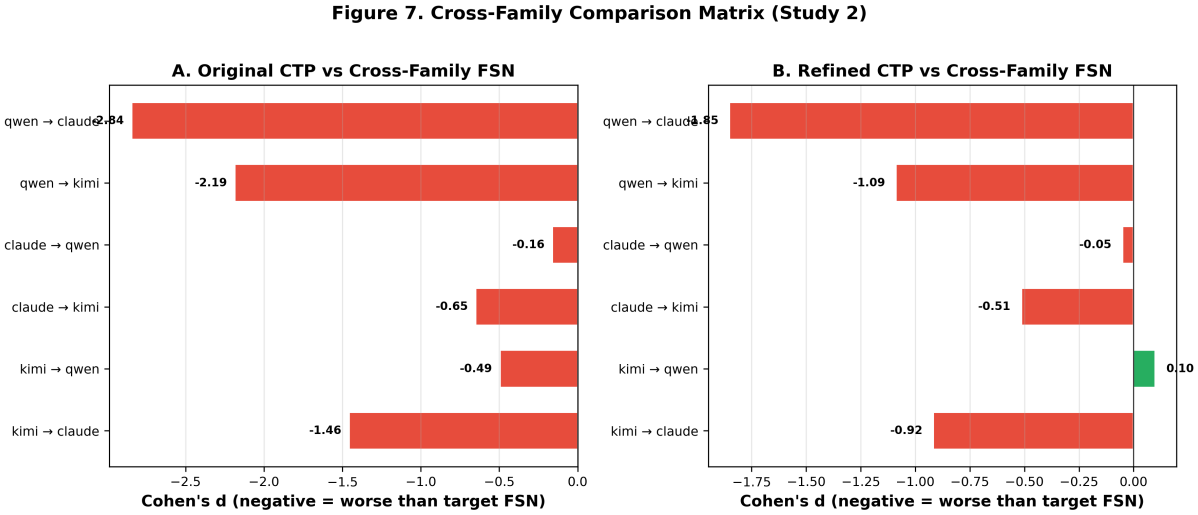


Figure 12: Cross-Family Comparison Matrix (Study 2).

depressed by near-zero prevalence. Conclusions on below-threshold codes are disclosed as medium reliability.

Of the six codes whose prevalence changed between original and refined (kimi, qwen), all six are *structural scaffolding* codes: `operational_parameters`, `structured_report`, `phased_timeline`, `tables_matrices`, `quantified_targets`, `granular_ops_detail`. No insight code (`diagnostic_reframe`, `named_concepts`, `assumption_test`, `concise_maxims`) moved. The convergence metric compares, for each of the 13 codebook codes, how far the original and refined lower-tier answers sit from the frontier answer's profile: a code counts as "toward" if refinement reduces that distance and "away" if it increases it. Refinement does not consistently move lower-tier outputs toward the frontier (kimi: toward 4 / away 6, net divergence; qwen: 5/4; claude: 0/2 with 11 codes flat — claude's originals were already at the codebook ceiling). Where outputs move away, refinement is adding structure the frontier answers do not carry — overshooting on coverage rather than gaining insight density.

The residual frontier edge is precisely located. The only κ-reliable frontier-vs-refined difference ≥ 0.30 is `concise_maxims`: qwen frontier 1.00 vs. refined 0.55 (gap 0.45). The

medium-reliability `redundancy_bloat` code is 0.00 in every frontier sample and 0.65–0.95 in every refined sample. Claude's original answers are already at the codebook ceiling (≥ 0.90 on 11 of 13 codes); the number of changed codes between original and refined is zero — a saturation mechanism that explains the 8% gap recovery in Section 4.2.1.

> **Strategic Implication:** Orchestration is a coverage play — it adds structure, length, and completeness. The frontier's advantage is a density play — compressed insight with zero waste. Add-only refinement cannot manufacture density, which explains why lower-tier teams approach but do not surpass frontier quality.

A note on replication scope. Unlike Study 1 and Study 3, whose headline claims were each stress-tested on fresh problems in a pre-registered replication (Section 4.1.7, Section 4.3.1), Study 2 was not carried into a confirmatory wave. Its two findings therefore rest on the original locked sample: the refinement ceiling is a pre-registered gate result — the G2 superiority test failed mechanically in all three families (Section 4.2.1) — and the coverage-versus-density mechanism is a mixed-methods coding result ($\kappa = 0.824$; Section 4.2.2). We report Study 2 as a *mechanistic explanation* of the Study 1 ceiling: well-supported on its own sample, but not yet independently replicated. The natural next step — a *subtractive* refiner permitted to cut as well as add — is named in Section 8.

Figure 5. Code Prevalence Heatmap: Original, Refined, and Frontier (Study 2)

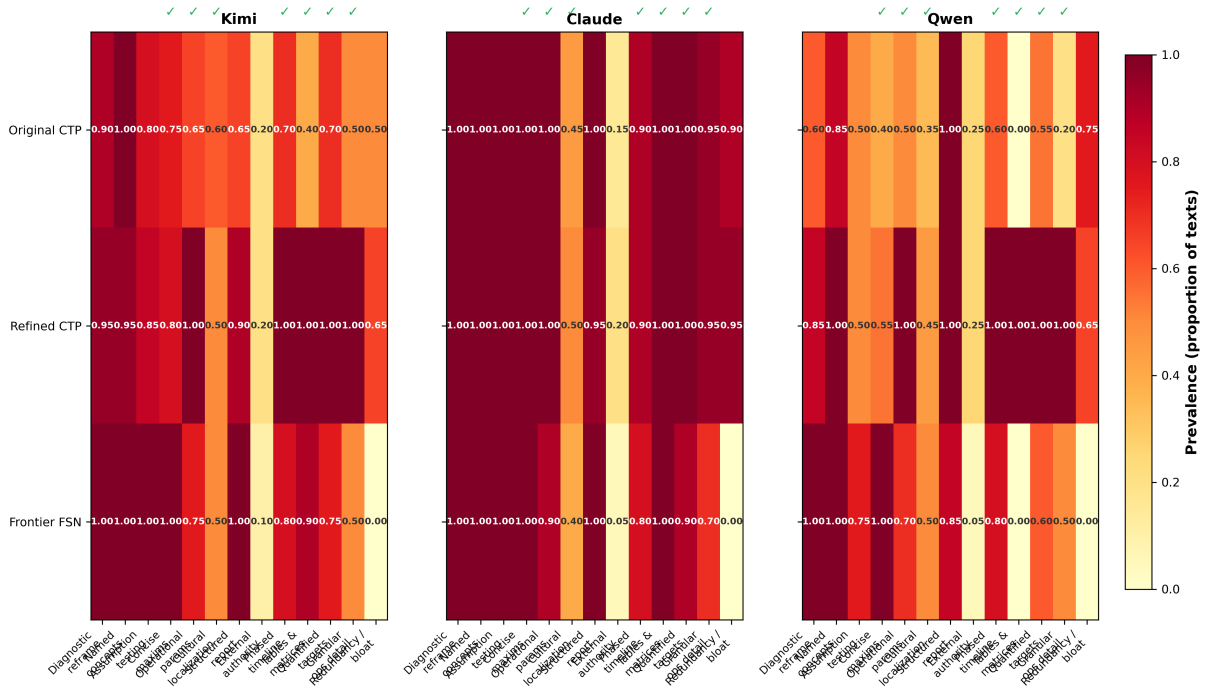


Figure 13: Code Prevalence Heatmap: Original, Refined, and Frontier (Study 2).

4.3 Study 3: Who Should Design the Pipeline?

Design. Every pipeline in Studies 1–2 was *human-designed*: the builder → critic → refiner structure was our fixed research instrument (the registered-but-unexecuted A7 arm isolated exactly this variable). Study 3 (Protocol 13; operative pre-registration OSF bwc93

— an initial v1 registration was superseded by this instrument amendment and withdrawn — with attrition clarification 639eq; all registrations timestamped before the corresponding computation) executes A7 as a stand-alone study. Using DeeWaanAI's production auto-designer, the model receives only the problem text and authors a bespoke multi-step pipeline — free choice of roles, step count, and critic-gated if/while feedback loops — which is then executed verbatim by the cheap tier. Two arms per family: **D-SELF** (the cheap sibling designs *and* executes) and **D-FRONT** (the frontier flagship designs; the cheap sibling executes). The 19 confirmatory problems × 3 replications × 2 arms × 5 families = 570 planned runs; 544 completed, with 26 disclosed attrition events concentrated in the Kimi lane (21 designer-stage failures — the cheap designer returned empty or truncated JSON on the production design prompt — plus 4 execution failures and 1 Claude designer failure). Solutions were judged by the same frozen three-judge panel under the three verbatim prompt variants (4,896 unique judgments). The primary contrast (A) tests H-D1: D-FRONT vs. D-SELF on Novelty, paired within family × problem, Holm-adjusted across families; contrast B tests H-D2, both arms against the frozen human-designed C-T-N baseline; an author-requested contrast informs H-D3, testing both arms against the frozen frontier-solo (F-S-N) baseline, and a cost analysis prices every run at the frozen price vector.

Contrast A: authorship barely matters. Table 5 reports the primary contrast. Frontier-designed pipelines are statistically indistinguishable from self-designed ones in four of five families; the lone significant effect favors frontier authorship for Claude ($d = 0.78$, $p_{\text{Holm}} = 0.014$). Pooled across families, the authorship gap on Novelty is 4.03 vs. 3.99 — under 0.1 scale points.

Table 5: Study 3 primary contrast: D-FRONT vs. D-SELF on Novelty, paired within family × problem ($n = 19$ pairs), Holm-adjusted across the five families.

Family	Δ (D-FRONT − D-SELF)	t	d	p_{raw}	p_{Holm}
Claude	+0.136	3.00	0.78	0.003	**0.014**
GPT-5.6	+0.049	1.77	0.22	0.078	0.310
Qwen	+0.021	0.73	0.10	0.463	1.0
Gemini	+0.021	0.55	0.08	0.582	1.0
Kimi	−0.038	−0.64	−0.12	0.522	1.0

Contrast B: both arms beat the human design. Against the frozen human-designed C-T-N pipeline, model-authored pipelines win in *all five families on both arms* (D-SELF $d = 0.15$–0.67; D-FRONT $d = 0.24$–1.33 on Novelty). The auto-designer also discovers structure we did not hand it: 98.9% (D-SELF) and 100% (D-FRONT) of designs include at least one critic-gated if/while feedback loop, averaging 5.3–5.4 designed steps and 5.8–6.2 executed steps after gate loops — richer than our fixed three-step instrument. Cheap designers iterate more at the design stage (D-SELF consumed 34.8M tokens vs. 24.3M for D-FRONT across the study).

Contrast vs. frontier solo: orchestration is family-specific, not general. Pooled, the arms are at parity with frontier solo (D-SELF $d = -0.22$; D-FRONT $d = -0.05$ on Novelty) — but the pooled figure hides the story (Table 6). Cheap auto-designed pipelines *beat* their own family's frontier flagship running solo for Kimi ($d = +0.85/+0.74$, $p_{\text{Holm}} = 0.006/0.013$), Qwen ($d = +0.55/+0.61$), and Gemini ($d = +0.46/+0.51$, n.s. after Holm); they tie GPT-5.6-solo; and they lose catastrophically to Claude-opus solo

($d = -3.15/-2.84$). By domain, D-FRONT meets or beats frontier solo in strategy, marketing, HR, and product, and loses in finance and operations — but this domain pattern does not survive within-family conditioning: the moderator is the *family* (how high the solo frontier sits), not the task. Frontier authorship does not rescue the losing families: where the cheap executor is the bottleneck, a frontier-designed plan cannot compensate.

Table 6: Study 3 economics: cost per completed solution (USD, frozen price vector v2026-08-23) and Novelty effect vs. frontier solo (F-S-N). ΔQ = Cohen's d on Novelty vs. F-S-N. Positive ΔQ at cost ratio < 1 = strictly dominant orchestration.

Family	**F-S-N cost**	**D-SELF**		**D-FRONT**	
		cost (×F-S-N)	ΔQ	cost (×F-S-N)	ΔQ
Gemini	\$0.043	\$0.010 (0.24×)	+0.46	\$0.025 (0.59×)	+0.51
Qwen	\$0.020	\$0.040 (2.0×)	+0.55	\$0.046 (2.3×)	+0.61
Kimi	\$0.107	\$0.377 (3.5×)	+0.85	\$0.274 (2.6×)	+0.74
GPT-5.6	\$0.180	\$0.729 (4.0×)	−0.07	\$0.512 (2.8×)	+0.13
Claude	\$0.266	\$0.438 (1.7×)	−3.15	\$0.668 (2.5×)	−2.84

Figure 14: Study 3 family × domain landscape: the Novelty effect of cheap auto-designed pipelines versus frontier solo (F-S-N) across the five families and six business domains. The binding moderator is the *family* — how high its solo frontier ceiling sits — not the task domain: the domain pattern does not survive within-family conditioning (Section 4.3.1).

Decision rule. Study 3 converts orchestration from a blanket recommendation into a routing table. For Gemini, cheap auto-designed pipelines are strictly dominant — better *and* 2–4× cheaper than frontier solo. For Qwen and Kimi, orchestration buys the largest quality gains in the study at 2–3.5× solo cost (which remains low in absolute terms). For Claude, no cheap-executed pipeline approaches the flagship: buy solo. For GPT-5.6, quality is a tie and solo is cheaper. And because authorship effects are negligible (Contrast A), when orchestration is indicated the cheap model can design for itself — the frontier design fee buys nothing, except where the cheap designer itself is fragile (21 of 57 Kimi self-design attempts — 37% — failed at the design stage, making frontier- or human-authored templates the safer default for that family). Section 4.3.1 stress-tests this table on new problems: the extremes (Kimi, Claude) transport, the mid-tier rows do not, so the table should be read as an illustration of the *method* rather than a durable lookup.

> **Key Takeaway:** who *designs* the pipeline does not matter — models author better pipelines than our human instrument, and cheap models author as well as frontier ones. Whether to *use* a pipeline depends on the family: orchestration is a substitute for tier where the frontier ceiling is low (Gemini, Qwen, Kimi) and a dominated cost where it is high (Claude).

4.3.1 Out-of-sample confirmation: does family moderation generalize? (Protocol 16)

Design. Study 3 estimated family-specific orchestration effects on the 19 confirmatory problems. Protocol 16 (pre-registered OSF zkm8v, registered and timestamped before

any generation) tests whether those estimates transport to *new* problems. We ran the D-SELF arm only — pre-declared, since Study 3 Contrast A found authorship effects negligible in 4/5 families — on 30 fresh wave problems × 3 replications × the same five core families = 450 planned runs; 441 completed with 9 disclosed attrition events (Gemini 90/90, Claude 89/90, Qwen 89/90, GPT-5.6 88/90, Kimi 85/90; feasibility gate G1 passed for every family). The comparator is the *frozen* wave frontier-solo (F-S-N) baseline — zero new frontier-solo generation, so the test is a pure transport test of Study 3's effect sizes. Solutions were judged by the same frozen three-judge panel under the three verbatim prompt variants (3,963/3,969 valid unique judgments; 6 judge calls returned persistently unparseable content on three Claude solutions and are disclosed as 0.15% judged-missing attrition). The analysis script was frozen and hash-locked (SHA-256 recorded in the lab manifest) *before* judging began. Five binding gates translate the Study 3 point estimates into directional predictions: R1 Kimi $d > 0$; R2 Qwen $d > 0$; R3 Gemini $d > 0$; R4 GPT-5.6 equivalence (TOST, 90% CI on d within ± 0.30); R5 Claude $d < -1.0$. The headline claim "family moderation replicates" requires R1 $\wedge$ R2 $\wedge$ R3 $\wedge$ R5.

Results: the extremes replicate; the middle does not. Table 7 reports the gate outcomes. Two of five gates pass, and they are the two *extreme* families: Kimi's cheap auto-designed pipelines again beat frontier solo, by an even larger margin than predicted ($d = +1.25$ vs. $+0.85$ predicted, one-sided $p < 10^{-4}$), and Claude's again lose badly ($d = -1.90$ vs. -3.15 predicted, well past the -1.0 threshold). The three middle families fail: Qwen's predicted gain vanishes ($d = -0.23$), and Gemini and GPT-5.6 *reverse sign* — their self-designed pipelines score significantly *below* frozen frontier solo on the new problems (Gemini $d = -0.83$; GPT-5.6 $d = -0.88$, 90% CI $[-1.32, -0.43]$, so the R4 equivalence gate fails in the negative direction, not toward zero). The headline conjunction is therefore **not replicated** (2/5 gates).

Table 7: Study 3 confirmatory replication (Protocol 16): binding gate outcomes. d = Cohen's d on Novelty, D-SELF vs. frozen wave F-S-N, problem-level Welch tests, $n = 30$ problem pairs per family. Predictions are the Study 3 point estimates.

Gate	Family	Predicted	Observed d	90% CI	Verdict
R1	Kimi	$d > 0$ (+0.85)	+1.25	$[+0.79, +1.72]$	**PASS**
R2	Qwen	$d > 0$ (+0.55)	−0.23	$[-0.66, +0.19]$	fail
R3	Gemini	$d > 0$ (+0.46)	−0.83	$[-1.28, -0.39]$	fail (reversed)
R4	GPT-5.6	$\lvert d \rvert < 0.30$	−0.88	$[-1.32, -0.43]$	fail (negative)
R5	Claude	$d < -1.0$ (−3.15)	−1.90	$[-2.41, -1.38]$	**PASS**

Interpretation. Family moderation is real but not a stable lookup table. The qualitative ordering — orchestration substitutes for tier where the frontier ceiling is low and is a dominated cost where it is high — survives at its extremes (Kimi, Claude), where effects are large. But for mid-tier families the sign and magnitude of the orchestration effect are problem-set-dependent: point estimates from 19 problems did not transport to 30 new problems drawn from the same generator. Combined with fast model turnover (the underlying model versions themselves change within months), any static per-family routing table is obsolete on arrival. The correct deliverable is therefore not a table but a *method*: cheap-tier benchmarking-as-routing — test the user's actual model on the user's actual problem type at trivial cost (a D-SELF probe run costs \$0.01–\$0.73 per solution; Table 6) and route on the result.

Why we did not extend to new families. A natural next step — repeating the study on the five additional families used in the wave replication — was considered and deliberately not registered. The scientific question it would answer (does family moderation exist for new families?) is already answered affirmatively by Studies 1–3; and Protocol 16's central finding is that mid-tier effect sizes do not transport even *within* known families across problem sets. Five additional families would therefore add point estimates subject to the same non-transport, at roughly 450 runs and 4,000 judgments of cost, without changing the methodological conclusion. We flag a bounded generalization probe on new families as future work, positioned as a product-method validation (benchmarking-as-routing) rather than a confirmatory experiment.

> **Key Takeaway:** the Study 3 extremes replicate on new problems — cheap pipelines beat frontier solo where the frontier ceiling is low (Kimi, $d = +1.25$) and lose where it is high (Claude, $d = -1.90$) — but mid-tier effects do not transport across problem sets. Orchestration routing must be measured per deployment, not looked up: benchmark-as-routing, not a routing table.

4.4 Summary of Key Findings

5 Discussion

5.1 A Decision Framework for AI Investment

The results support a contingency framework for AI investment rather than a universal rule. For budget-constrained firms, **invest in the lower tier first**: a single call from a cheaper model (C-S-N) achieves 98% of frontier quality at 23% of the cost (T-EGR 6.15) — the highest-return configuration in the study. **Add the three-step pipeline only when coverage is the target** (timelines, matrices, quantified plans): the cheaper team (C-T-N) reaches 98.8% of frontier quality at 88% of the cost (T-EGR 1.49). **Do not orchestrate at the frontier tier**: frontier team cells spend 3.5× the frontier-solo cost for +0.024 points of quality (T-EGR 0.27), making them the worst dollar in the experiment. **Use a low-cost refinement pass as a risk-mitigation strategy** when absolute quality parity is the target: the refined kimi point reaches parity at 78% of F-S-N's cost, the lowest-cost route to that milestone among the configurations tested — noting that parity here is on pooled means, while the paired within-family comparison remains negative (Section 4.2.1). **Avoid personas**: they degrade quality sharply while reducing cost only marginally — a value-destroying configuration.

Study 3 adds a family-conditional override to this framework (Section 4.3, Table 6): where the deployment stack is Gemini, the auto-designed cheap pipeline is strictly dominant over frontier solo (better novelty at 0.24× the cost); for Qwen and Kimi the pipeline is the quality-maximizing choice at 2–3.5× solo cost; for Claude the solo flagship remains untouchable and pipelines are a dominated expense; and in all cases the design fee is better spent on execution than on authorship — self-designed pipelines match frontier-designed ones in four of five families.

5.2 The Density Ceiling: Why Additive Pipelines Hit a Wall

The qualitative mechanism (Section 4.2.2) explains the pattern. Orchestration is compositional — add a critic, add a refiner, add a section, add a table. Composition grows

Table 8: Summary of key findings across the three studies and their pre-registered replications.

Finding	Evidence	Section
Team structure helps lower-tier models more than frontier (H_1)	$d = -0.227$	Section 4.1.1
Frontier > cheaper overall (H_2)	$d = +0.294$	Section 4.1.1
Teams slightly beat solos on average (H_3)	$d = +0.117$	Section 4.1.1
Persona penalty on net (H_4, reversed)	$d = -0.691$	Section 4.1.1
Structure × persona interaction (H_5, n.s.)	$F = 1.23$, $p = 0.27$	Section 4.1.1
Persona-loaded cells Pareto-dominated	cost–quality plane	Section 4.1.2
CTP < FSN in every family	paired comparisons	Section 4.1.3
Refinement recovers up to 64% of gap (8% where gap already small); G2 rejected	Protocol 09 tests (H2.1–H2.3)	Section 4.2.1
Orchestration buys coverage; frontier wins on density	$\kappa = 0.824$; 13 codes	Section 4.2.2
Pareto frontier stable across φ	{C-S-P, C-S-N, C-T-N, F-S-N, F-T-N}	Section 4.1.4
Replicated (wave): team effect, upgrade threshold, bimodal economics	W1/W2/W6 PASS, 10 families, $n = 2{,}418$	Section 4.1.7
Exploratory (gate FAILED in wave): domain contingency; capability-gap mechanism; marketing variance advantage	W3/W4/W5 FAIL	Section 4.1.6, Section 4.1.7
Exploratory (not wave-tested; wave excluded persona cells): persona novelty gain is solo-only; teams squash it	D45 battery	Section 4.1.6
Design authorship negligible (4/5 families); auto-designed > human-designed pipelines (5/5)	Study 3, Protocol 13	Section 4.3
Cheap auto-designed pipelines beat frontier solo for Gemini/Qwen/Kimi; lose for Claude; tie GPT	d up to +0.85; cost 0.24–3.5× solo	Section 4.3
Partially replicated: extremes transport to new problems (Kimi $d = +1.25$; Claude $d = -1.90$); mid-tier families do not — routing must be measured, not tabulated	Protocol 16, 2/5 gates, $n = 441$	Section 4.3.1

the answer but cannot compress it. The frontier's residual advantage sits exactly where additive composition fails: `concise_maxims` (compressed, quotable strategic principles) and zero `redundancy_bloat`. These are the outputs of a single model that has been selected or trained to produce dense prose. An add-only refiner, by design, cannot delete the boilerplate the team structure produced. The falsifiable prediction that follows is that a *subtractive* refiner — one permitted to cut as well as add — would narrow the density gap. This is the central future-work experiment implied by our results.

5.3 Contribution to the Business of AI Literature

Our findings extend the author's prior lineage in two directions. The EGR metric introduced in Hariri (2026b) measured the cost performance of *personas in single models*; T-EGR extends it to teams and pipelines, preserving the ratio form while re-basing on the frontier cell. The team-composition contingency identified on verifiable tasks in Hariri (2025b) reappears here on open-ended tasks, but with the sign reversed for personas (harmful rather than neutral on substance). The jagged-frontier finding of Dell'Acqua et al. (2023) is replicated as a *cell-level* result: lower-tier models inside the persona-free, coverage-oriented region beat their cost-matched frontier alternatives; cheaper models

Figure 20. Practitioner Contingency Rules

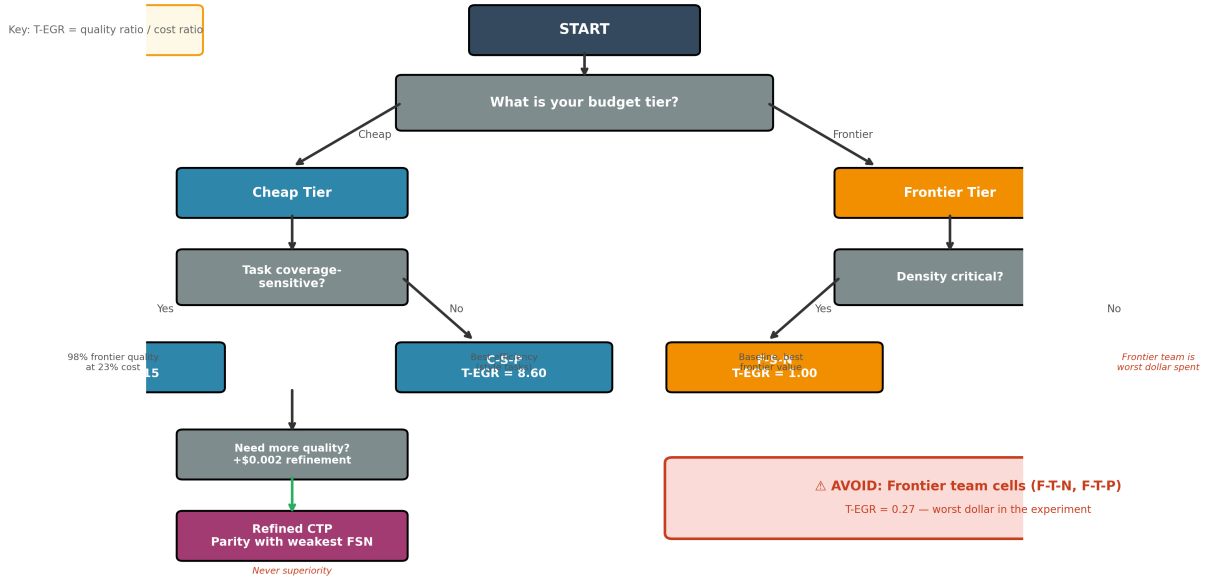


Figure 15: Practitioner contingency rules. Decision flowchart mapping budget tier and task type to the recommended orchestration cell, with expected T-EGR. Derived from the Section 5.1 contingency model.

pushed outside by persona-loading or by density demands fall below the frontier.

The test-time-compute prediction of Snell et al. (2024) is partially confirmed: more inference compute (team, refinement) improves the cheaper model, but only up to a parity ceiling, and only when priced in dollars does the tradeoff reveal its practitioner shape. Our contribution is the dollarization of the compute-optimal curve on open-ended tasks where no verifier exists.

5.4 Methodological Lessons for AI Deployment Research

Two lessons carry beyond this study. First, preregistered negative superiority results can be the spine of a paper, not its appendix. The G2 mechanical failure in Section 4.2.1 is as informative as any positive finding, and preregistration is what made it reportable as a primary result rather than as a null. Second, token-invariant costing — storing tokens and re-pricing through a frozen vector — is reusable infrastructure for the field: every reported dollar is reproducible by any reader who supplies an alternative price vector (Diewert, 1976; Sandve et al., 2013).

6 Managerial Implications

This section translates the empirical findings into decisions a budget owner can act on. Where Section 5.1 states the contingency framework as a research result, the guidance below is written as practice: a sequenced budget rule, an operating model, a risk register, and a sourcing decision. Every claim carries its dollar figure first and its supporting statistic second, and points to the quick-reference table (Table 9) rather than repeating it.

Table 9: Managerial quick-reference: recommended configuration by task profile.

Task profile	Cell	Quality	Cost	T-EGR	Override when...
Coverage-heavy (plans, timelines, matrices)	C-S-N	98%	23%	6.15	absolute quality non-negotiable → F-S-N
Density-critical (exec briefs, strategic principles)	F-S-N	100%	100%	1.00	budget tight → C-S-N at 98%
Budget-minimum, quality-adequate	C-S-P	92%	17%	8.60	novelty needed, substance traded
Refinement as risk mitigation	Refined C-T-P	∼100%	78%	—	never achieves paired superiority
Avoid	F-T-N	100.5%	353%	0.27	worst dollar in the experiment
Avoid	F-T-P	95.3%	365%	0.27	persona penalty on top of team overhead

6.1 The Investment Ladder: A Sequenced Budget Rule

The central managerial finding is that AI capability should be *purchased in a sequence*, not all at once. Spend down the following ladder and stop at the rung where the next dollar buys less quality than it costs:

1. **Default to a single lower-tier call (C-S-N).** It delivers 98% of frontier quality for 23% of the price (T-EGR 6.15) — the highest-return dollar in the study and the correct default for most open-ended knowledge work.
2. **Add the three-step team only when coverage is the deliverable (C-T-N)** — timelines, option matrices, quantified plans. It reaches 98.8% of frontier quality at 88% of the cost (T-EGR 1.49): still positive-return, but the marginal dollar now buys *coverage*, not quality.
3. **Buy a refinement pass as cheap insurance ($0.002).** It lifts a refined lower-tier answer to parity with the weakest frontier at 78% of frontier-solo cost — but it never achieves superiority. Treat it as a risk-mitigation rung, not a replacement rung.
4. **Reserve frontier solo (F-S-N) for density-critical output** — executive briefs and compressed strategic principles, where the residual frontier advantage actually lives (Section 4.2.2).
5. **Never orchestrate at the frontier tier (F-T-N).** A frontier team costs 3.5× the frontier-solo price (353% of cost) for negligible quality gain (T-EGR 0.27) — the worst dollar in the experiment.

Strategic implication. The budget question is not "frontier or cheap?" but "which rung, for which task?" A firm that routes coverage work to the lower tier and reserves the frontier for density captures most of the quality at a fraction of the cost; a firm that pays frontier prices for team orchestration overpays for output it could have bought cheaper.

6.2 An Operating Model: Route by Measurement, Not by Table

Study 3 changes how the ladder is operated. The economics of orchestration are *family-specific* (Section 4.3, Table 6): the auto-designed cheap pipeline strictly dominates frontier solo on a Gemini stack at 0.24× the cost, maximizes quality for Qwen and Kimi at 2–3.5×

solo cost, and is a dominated expense on Claude, where the solo flagship is untouchable. The managerial consequence is direct: **there is no universal routing table a firm can copy.** The operating model is a *benchmark-and-route* discipline — on a small held-out set of the firm's own tasks, measure the cheap-pipeline-versus-frontier-solo T-EGR for the specific model family in the stack, then route on the measured number.

Two further operating rules follow. First, **spend the design fee on execution, not authorship**: self-designed pipelines matched frontier-designed ones in four of five families, so a firm does not need to pay a frontier model to author its workflows. Second, **treat routing as a recurring procurement KPI**, re-measured whenever a model version or an API price changes — because the dollar figures that drive the ladder move with the market, even when the quality ordering does not.

6.3 Risk Register: What Can Go Wrong and How to Bound It

- **Persona risk.** Role/persona prompting degrades substance while reducing cost only marginally (Section 4.1.2) — a value-destroying configuration that is easy to adopt by accident because it "feels" more professional. *Control:* audit persona usage and strip it wherever substance, not style, is graded.
- **Density risk.** Orchestration buys coverage but not density, and a refinement pass closes most of the gap without ever beating the frontier (Section 4.2.2, Section 4.2.1). *Control:* do not expect a cheap additive pipeline to substitute for compressed, quotable insight; buy the frontier for that class of output.
- **Replication risk — honestly bounded.** The core economics replicated out-of-sample across 10 model families (Protocol 12), but the exploratory family-extremes did not transport cleanly (Protocol 16 passed 2 of 5 pre-registered gates). *Control:* treat the headline rules as bankable and the family-specific extremes as hypotheses to re-measure locally, not as guarantees. This is the payoff of pre-registration — we can tell the manager which findings survived a fresh bet and which did not.

6.4 Build-versus-Buy: A Capability-Sourcing Decision

Frontier models are *bought*: a fixed per-call cost for a given quality level. Orchestration pipelines are *built*: an upfront architectural investment that lowers the marginal cost per call. The build path pays off when (a) the task is coverage-sensitive, (b) the quality bar is "good enough" rather than "best available," and (c) the firm can absorb the engineering cost of maintaining a multi-step pipeline. When density is the binding constraint, the buy path (frontier solo) remains optimal.

The decision statistic is T-EGR: **build when the measured T-EGR on your own workload exceeds 1, buy when the frontier's density premium is the thing you actually need.** Critically, this is a *per-task-family* decision, not a firm-wide posture — the same organization should build cheap pipelines for coverage work and buy frontier calls for density work, and should expect the mix to shift as model prices and capabilities move.

7 Limitations

Six acknowledged limitations bound the claims. (1) Evaluation is LLM-only; no human panel was recruited. Our bias controls (blinding, order shuffle, disjoint judge families,

three prompt variants) follow the best current practice (Zheng et al., 2023; Verga et al., 2024) but do not substitute for human agreement on a subset. (2) The blended-φ price assumption is a disclosed limitation; we mitigate it with a pre-registered sensitivity panel and require cross-φ stability for any frontier-membership claim. (3) The task domain is English business strategy; generalization to code, scientific writing, or non-English domains is an empirical question. (4) Cross-family comparisons (e.g. kimi vs. claude refinement rates) are descriptive and confounded with family quality; only within-family comparisons support causal language. Two further disclosures — the small truncation audit and the team-cost retry gap — are stated but not corrected. On the latter: team intermediate outputs are excluded from judged answers by design (only the final answer is scored) but *are* included in cost; the sole cost understatement comes from failed step attempts whose retry tokens were not logged (110 of 950 completed team runs, a median true-to-recorded cost ratio of 1.69 on affected runs — roughly 8% of team-cell cost on average). The gap mildly *flatters* the team cells, but it is immaterial against the cost differences that drive the conclusions (teams cost $\sim$3.8$\times$ the solo call): no ordering can flip.

A fifth limitation is practical: our costs are API-list prices, not enterprise-negotiated rates. Firms with volume commitments or reserved-capacity agreements may face different absolute prices, though the *relative* rankings (lower tier $\approx$ 23% of frontier) are preserved as long as the price ratio between tiers remains in the 3–8$\times$ range observed across the five families at the time of the experiment. Similarly, pipeline latency — the wall-clock time of three sequential API calls versus one — is not priced here; latency-sensitive applications may prefer frontier solo even when the dollar cost favors the lower-tier pipeline.

A sixth limitation concerns stimulus sampling breadth. The confirmatory analysis rests on 19 business problems — below the 30–40 stimulus range recommended for maximal generalizability power (Westfall et al., 2014). Although leave-one-problem-out cross-validation (19 folds) confirmed sign stability of every reported effect (max $|\Delta d| = 0.02$), and the within-problem replication is dense (100 runs/problem, 5 families), the 19 problems do not exhaust the universe of possible business tasks. External validity — the claim that orchestration economics generalizes beyond these specific problems — is therefore bounded. Future work should replicate with larger, more diverse problem corpora spanning additional domains, languages, and task complexities.

A seventh limitation concerns the replication wave's execution environment (Section 4.1.7). Two provider-side events required disclosed handling: the Gemini lane ran after a billing upgrade resolved a free-tier daily request cap, and the Kimi frontier lane was completed via a Together.ai routing deviation after Moonshot credit exhaustion (D53) — the pinned model version was unchanged, but within-family routing homogeneity for Kimi was broken (cheap cells Moonshot-direct, frontier via Together; per-run route membership is auditable in the ledger). Separately, the pre-registered dual variance-component mixed model was numerically singular on the wave design matrix; a pre-declared fallback rule (hard-requiring bit-identical fixed effects) substituted a parsimonious random-effects structure (D52), so the W3a likelihood-ratio statistic carries a REML-comparability caveat.

An eighth limitation is specific to Study 3 (Section 4.3). Attrition was concentrated in one family: 37% of Kimi self-design attempts (21 of 57) failed at the *designer* stage (empty or truncated structured output on the production auto-design prompt), and total Kimi-lane attrition reached 22% (25 of 114, including a provider moderation block on one problem), so the Kimi cells rest on fewer observations ($n = 89$ of 114) and the designer-

fragility finding is itself confounded with provider routing (Moonshot vs. Together; D58). Study 3 also instantiates design authorship through a *single* production auto-designer at its default decoding settings; other designers, temperatures, or step caps might shift the authorship contrast, and our instrument deliberately disallows human-in-the-loop approval steps, so the results speak to fully automated pipelines only.

8 Future Directions

The first priority named in earlier drafts — an out-of-sample confirmatory replication wave — is now complete (Section 4.1.7); it promoted the core claims and rejected the exploratory upgrades. The second — pipeline design authorship — is likewise now complete (Study 3, Section 4.3): it executed the registered A7 arm and found authorship effects to be negligible, with orchestration economics family-specific. Two candidates from the original roadmap are now closed by the results themselves: **loop control** (Protocol 14) is superseded because Study 3 found critic-gated loops in 98.9–100% of auto-designed pipelines yet outcomes remained family-determined, and **static routing** (Protocol 15) is superseded because Protocol 16 showed mid-tier effects do not transport across problem sets — routing is a measured per-deployment discipline (Section 4.3.1), not a learnable lookup. Four experiments follow directly. First, a **bounded generalization probe** on new model families, positioned as a validation of the benchmarking-as-routing method rather than a confirmatory experiment (Section 4.3.1). Second, a **subtractive refiner** study that is permitted to cut as well as add: this isolates the density mechanism identified in Section 4.2.2. Third, a **human-panel replication** on a subset of runs to bound LLM-judge bias in our specific design. Fourth, a **second task domain** (code generation, scientific writing) to test whether the coverage-vs-density mechanism generalizes beyond business strategy.

9 Conclusion

For organizations deploying LLMs for knowledge work, three investment rules emerge from four pre-registered empirical bases: Study 1 (1,900 runs), its Protocol 12 replication wave (2,418 analyzed runs), Study 3 (544 completed runs), and the Protocol 16 replication (441 completed runs) — together spanning 10 model families and 30 fresh problems. **First**, at the lower-tier end, model-tier substitution works: a single call from a lower-tier model delivers 98% of frontier quality at 23% of the cost (T-EGR 6.15) — the highest-return configuration tested; the three-step team variant adds coverage at 88% of the cost (T-EGR 1.49). **Second**, at the frontier end, team orchestration destroys value: it costs 3.5× the frontier-solo price for negligible quality gain (T-EGR 0.27). **Third**, a low-cost refinement pass ($0.002) recovers most of the frontier gap but never beats it — a bounded-improvement result that makes it a viable risk-mitigation strategy, not a replacement strategy. The residual frontier advantage is density (compressed insight, zero waste), not coverage — a capability gap that add-only pipelines cannot bridge. Personas are value-destroying: they degrade quality sharply ($d = -0.691$) while reducing cost only marginally. The governing principle is contingency, not universality: match the right pipeline architecture to the right model tier for the right task, and invest in refinement before investing in a frontier team.

The broader implication is a build-versus-buy framework for AI capability. Frontier

models are *bought* — a fixed per-call cost for a given quality level. Orchestration pipelines are *built* — an upfront architectural investment that lowers marginal cost per call. Our results show that the build path pays off when (a) the task is coverage-sensitive, (b) the quality bar is "good enough" rather than "best available," and (c) the firm can absorb the engineering cost of maintaining a multi-step pipeline. When density is the binding constraint — when the output must compress insight into minimal prose — the buy path (frontier solo) remains optimal. The contingency is not a limitation of the study; it is the finding.

For the field, the contribution is a dollarized, pre-registered, cost-matched empirical test of the substitution claim that sits at the heart of the multi-agent AI industry. The claim is partially true: structure does substitute for tier, but only up to a ceiling set by density. We offer this ceiling, and the contingency rules that surround it, as one empirical data point toward a more rigorous economics of AI orchestration.

A Executive Summary for Practitioners

Decision context. A firm that needs LLM-generated knowledge work can either (a) pay for a frontier model or (b) use lower-tier models, optionally orchestrated into a multi-step pipeline. Which dollar buys more quality?

Bottom line. Across 1,900 runs, five model families, and 19 business problems: a single call from a lower-tier model delivers **98% of frontier quality at 23% of the cost** (T-EGR 6.15). Orchestrating that model into a three-step builder–critic–refiner pipeline adds coverage for about 3.8× that cost (T-EGR 1.49). Orchestrating a frontier model costs **3.5× more for negligible gain**. A $0.002 refinement pass recovers most of the remaining gap but never beats the frontier.

Three rules for AI budget allocation:

1. **cheaper tier, single call** → highest ROI (T-EGR 6.15). Add the three-step pipeline only when coverage is the target — timelines, matrices, quantified plans — at ∼3.8× the solo cost (T-EGR 1.49).
2. **Frontier tier (solo)** → best absolute quality per dollar when density matters (compressed insight, zero boilerplate).
3. **Never add personas as a quality lever** — they degrade judged quality sharply ($d = -0.691$) while reducing cost only marginally.

Caveat. Cheaper configurations buy *coverage* (more structure, more length); frontier models win on *density* (insight per paragraph). If the task rewards concise strategic principles, the frontier is worth the premium. If the task rewards completeness, the lower-tier configuration is the better bet.

From findings to product. This research was conducted on, and now feeds back into, the DeeWaanAI orchestration platform. Three findings have already changed the product: (1) the persona penalty ($d = -0.691$) removed personas as a default quality lever; (2) Study 3 established that pipeline *design authorship* is a negligible factor, so the platform's auto-designer lets the cheap model design for itself rather than billing a frontier design pass; and (3) Protocol 16's partial replication converted the static per-family routing table into *benchmarking-as-routing* — a cheap per-deployment probe (test the user's model on the user's problem type, then route), because mid-tier effects do not transport across problem sets. The loop is deliberate: the platform supplies the research instrument, the pre-registered findings supply the product's routing logic, and the disclosure in Section 3.9 keeps the relationship transparent.

B Protocol and Arm Registry

A **protocol** is a pre-registered experiment plan frozen before its data are generated; an **arm** is one experimental condition within a protocol (Section 3). Table 10 lists every numbered protocol and amendment in the research program with its role and status; Table 11 defines every arm label used in the text.

Table 10: Protocol registry. "Frozen" = time-stamped before any data for that question were generated. OSF GUIDs are mapped in Table 2.

Protocol	Role	S
01	Study 1 factorial pre-registration (H_1–H_5)	fr
02	Qualitative analysis (deductive codebook)	fr
03	Sequential analysis (O'Brien–Fleming interims)	fr
04	Cost-effectiveness analysis (T-EGR, Pareto)	fr
05–08	Amendments A1–A4 (pilot corrections; extended interim-2; exploratory statistics; inductive qualitative)	fr
09	Study 2: refinement probe	fr
10	Study 2: qualitative coding	fr
11	Study 2 extension: refinement (GPT, Gemini; cross-family refiner)	fr
12	Confirmatory replication wave (30 fresh problems, 10 families)	C
13	Study 3: pipeline design authorship (D-SELF vs. D-FRONT)	C
14	Loop control (critic-gated cycles)	d
15	Static routing from problem text	d
16	Study 3 confirmatory replication (family-moderation transport)	C

Table 11: Arm registry. The eight factorial cells are tier (C = cheaper, F = frontier) × structure (S = solo, T = team) × persona (N = none, P = persona). A5–A8 are mechanism arms registered in Study 1; D-SELF/D-FRONT are the Study 3 arms.

Arm	Definition	Status
C-S-N	cheaper solo, no persona	executed (Study 1 + wave)
C-S-P	cheaper solo, persona	executed (Study 1)
C-T-N	cheaper team (builder → critic → refiner), no persona	executed (Study 1 + wave)
C-T-P	cheaper team, persona	executed (Study 1)
F-S-N	frontier solo, no persona (baseline)	executed (Study 1 + wave)
F-S-P	frontier solo, persona	executed (Study 1)
F-T-N	frontier team, no persona	executed (Study 1 + wave)
F-T-P	frontier team, persona	executed (Study 1)
A5	critic-gated WHILE loop on the cheap team	registered; never executed in Study 1
A6	cost-matched frontier-solo baseline	registered; never executed in Study 1
A7	adaptive per-problem pipeline designer	registered; executed as Study 3
A8	shuffled persona-to-role mapping	registered; never executed in Study 1
D-SELF	cheap sibling designs *and* executes the pipeline	executed (Study 3 + Protocol 16)
D-FRONT	frontier flagship designs; cheap sibling executes	executed (Study 3)

References

Albers, C., & Lakens, D. (2018). When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of Experimental Social Psychology*, *74*, 187–195. doi:10.1016/j.jesp.2017.09.004

Borenstein, M., Hedges, L. V., Higgins, J. P. T., & Rothstein, H. R. (2009). *Introduction to Meta-Analysis*. Wiley. doi:10.1002/9780470743386.

Brown, M. B., & Forsythe, A. B. (1974). Robust tests for the equality of variances. *Journal of the American Statistical Association*, 69(346), 364–367. doi:10.1080/01621459.1974.10482955.

Brynjolfsson, E., Li, D., & Raymond, L. R. (2023). *Generative AI at Work* (NBER Working Paper No. 31161). doi:10.3386/w31161

Camerer, C. F., Dreber, A., Holzmeister, F., Ho, T.-H., Huber, J., Johannesson, M., … Wu, H. (2018). Evaluating the replicability of social science experiments in *Nature* and *Science* between 2010 and 2015. *Nature Human Behaviour*, 2(9), 637–644. doi:10.1038/s41562-018-0399-z

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, *20*(1), 37–46. doi:10.1177/001316446002000104

Dell'Acqua, F., McFowland III, E., Mollick, E. R., Lifshitz-Assaf, H., Kellogg, K., Rajendran, S., Krayer, L., Candelon, F., & Lakhani, K. R. (2023). *Navigating the jagged technological frontier: Field experimental evidence of the effects of AI on knowledge worker productivity and quality* (HBS Working Paper No. 24-013). doi:10.2139/ssrn.4573321

Dietterich, T. G. (2000). Ensemble methods in machine learning. In *Multiple Classifier Systems (MCS 2000), Lecture Notes in Computer Science, 1857*, 1–15. Springer. doi:10.1007/3-540-45014-9_1.

Dubois, T., Liang, P., & Hashimoto, T. B. (2024). Length-controlled AlpacaEval: A simple way to debias automatic evaluators. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*. arXiv:2404.04475.

Diewert, W. E. (1976). Exact and superlative index numbers. *Journal of Econometrics*, *4*(2), 115–145. doi:10.1016/0304-4076(76)90009-9

Hansen, L. K., & Salamon, P. (1990). Neural network ensembles. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(10), 993–1001. doi:10.1109/34.58877.

Hariri, Y. (2025a). *LLM-Creativity* [dataset and paper]. Zenodo. doi:10.5281/zenodo.17407392

Hariri, Y. (2025b). *LLM Team Composition Paradox* [dataset and paper]. Zenodo. doi:10.5281/zenodo.17490945

Hariri, Y. (2026b). *Doing More with Less: How Cultural Personas Bridge the Cost–Performance Gap in LLMs*. Zenodo. doi:10.5281/zenodo.19533619

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70. doi:10.2307/4615733.

Kendall, M. G., & Babington Smith, B. (1939). The problem of m rankings. *The Annals of Mathematical Statistics*, 10(3), 275–287. doi:10.1214/aoms/1177732186.

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for *t*-tests and ANOVAs. *Frontiers in Psychology*, *4*, 863. doi:10.3389/fpsyg.2013.00863

Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Shi, S., & Tu, Z. (2024). Encouraging divergent thinking in large language models through multi-agent debate. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing* (pp. 17889–17904). doi:10.18653/v1/2024.emnlp-main.992

Open Science Collaboration. (2015). Estimating the reproducibility of psychological science. *Science*, *349*(6251), aac4716. doi:10.1126/science.aac4716

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *PNAS*, *115*(11), 2600–2606. doi:10.1073/pnas.1708274114

Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science*, *381*(6654), 187–192. doi:10.1126/science.adh2586

Sandve, G. K., Nekrutenko, A., Taylor, J., & Hovig, E. (2013). Ten simple rules for reproducible computational research. *PLoS Computational Biology*, *9*(10), e1003285. doi:10.1371/journal.pcbi.1003285

Shinn, N., Cassano, F., Berman, E., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, pp. 8634–8652.

Shrout, P. E., & Fleiss, J. L. (1979). Intraclass correlations: Uses in assessing rater reliability. *Psychological Bulletin*, 86(2), 420–428. doi:10.1037/0033-2909.86.2.420.

Snell, C., Lee, J., Xu, K., & Kumar, A. (2024). Scaling LLM test-time compute optimally can be more effective than scaling model parameters. arXiv:2408.03314. doi:10.48550/arXiv.2408.03314

Steegen, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science*, *11*(5), 702–712. doi:10.1177/1745691616658637

Tomlinson, B., Black, R. W., Patterson, D. J., & Torrance, A. W. (2024). The carbon emissions of writing and illustrating are lower for AI than for humans. *Scientific Reports*, *14*, 3732. doi:10.1038/s41598-024-54271-x

Verga, P., Hofstätter, S., Althammer, S., Su, Y., Piktus, A., Arkhangorodsky, A., Xu, M., White, N., & Lewis, P. (2024). Replacing judges with juries: Evaluating LLM generations with a panel of diverse models. arXiv:2404.18796. doi:10.48550/arXiv.2404.18796

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., & Zhou, D. (2023). Self-consistency improves chain of thought reasoning in language models. In *Proceedings of the Eleventh International Conference on Learning Representations* (ICLR 2023).

Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E., Le, Q., & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, pp. 24824–24837.

Westfall, J., Kenny, D. A., & Judd, C. M. (2014). Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General*, *143*(5), 2020–2045. doi:10.1037/xge0000014

Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T. L., Cao, Y., & Narasimhan, K. (2023). Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, Datasets and Benchmarks Track.